

# Digital Wave Mechanics

Dylon La Rue

## Digital Wave Mechanics

*From Newton's Prism to Standing Waves  
in Finite Golden Fields*

Dylon La Rue & James Lockwood

Independent Research · Las Vegas, NV

WMR2 — June 2026 | *math.NT* · *math-ph* · *math.GR*

**Epistemic Note.**

A *theorem* is an algebraic or analytic statement proved from stated hypotheses. A *verified computation* is a finite calculation whose inputs and outputs are given explicitly and whose replication requires only modular exponentiation or the displayed script. A *physical interpretation* or *structural analogy* is a map from the algebraic object to physical language. Physical interpretations are not promoted to physical laws unless a measurement protocol, observable, null model, and falsification criterion are stated.

This separation is not cosmetic. The identity  $1 + \rho + \rho^2 = 0$  is a theorem in both  $\mathbb{C}$  and finite fields of suitable characteristic. The statement that this identity captures the center algebra of  $SU(3)$  is a group-theoretic theorem. The stronger statement that a finite-field model reproduces full Algebraic confinement would require an analog of gauge fields, Wilson loops, running coupling, asymptotic freedom, and an area law. That stronger statement remains an open physical problem residing beyond this epistemic boundary.

## 0.1 Computational Methods and Reproducibility

The computational pipeline verifies the algebraic claims in this chapter through three independent procedures, all operating in exact integer arithmetic over finite fields.

**Galois Splitting Verification.** The golden polynomial  $X^2 - X - 1$  is tested for splitting over all odd primes  $p < 10,000$  (excluding  $p = 5$ , which is ramified). For each prime  $p$ , the script computes the Legendre symbol  $\left(\frac{5}{p}\right)$  via modular exponentiation:  $5^{(p-1)/2} \bmod p$ . Splitting is confirmed when this equals  $+1$ , and the result is cross-checked against the theoretical criterion  $p \equiv 1 \text{ or } 4 \pmod{5}$  from quadratic reciprocity. Every prime is asserted to satisfy both the computational and theoretical conditions. The implementation uses `sympy.primerange()` for sieving and Python's native `pow(a, b, m)` for modular exponentiation, which internally uses the binary method ( $O(\log n)$  multiplications).

**Order Certification.** For each golden split prime  $p$ , the multiplicative orders  $\text{ord}_p(\varphi)$  and  $\text{ord}_p(\bar{\varphi})$  are certified by the method of Lemma 0.4: (1) verify  $a^N \equiv 1 \pmod{p}$  by direct computation, then (2) for each prime divisor  $q \mid N$ , verify  $a^{N/q} \not\equiv 1 \pmod{p}$ . This is implemented with the standard library's `pow()` and `sympy.factorint()` for prime factorization of the candidate order. The orbit classification (golden, conjugate, or full-order) is determined by comparing  $\text{ord}_p(\varphi)$  against the divisors of  $p - 1$ .

**Adelic Chronometry.** The Lockwood chronometric triangle  $(116, 138, 144)$  is verified by computing  $a \bmod p$  for each side  $a \in \{116, 138, 144\}$  at all three gauge primes  $p \in \{229, 181, 139\}$ , then classifying the residues as golden-orbit or conjugate-orbit elements by checking membership in  $\langle \varphi \rangle$  vs.  $\langle \bar{\varphi} \rangle$ . The Heron area-squared  $s(s-a)(s-b)(s-c) = 55,414,535$  is verified by exact integer arithmetic. The implementation uses the `fractions` module for exact rational arithmetic and `mpmath` for high-precision verification of the Lockwood log-time kernel  $S(x) = 1/(1 + Y(x))$ .

**Software Environment.** Python 3.10+. Standard library: `math`, `fractions`, `sys`. External: `sympy`  $\geq 1.12$  (prime sieving, factorization), `mpmath`  $\geq 1.3$  (arbitrary-precision real arithmetic). No numerical optimization, no machine learning, no floating-point approximations in the core pipeline. All modular arithmetic is exact.

## 0.2 Arithmetic conventions

All congruences are taken in the ring indicated by the modulus. When  $p$  is prime,  $\mathbb{F}_p$  denotes the finite field with  $p$  elements and  $\mathbb{F}_p^\times$  denotes its multiplicative group, cyclic of order  $p - 1$ . For  $a \in \mathbb{F}_p^\times$ ,  $\text{ord}_p(a)$  denotes the least positive integer  $n$  such that  $a^n \equiv 1 \pmod{p}$ .

The polynomial used throughout is  $f(X) = X^2 - X - 1$ . If  $f$  splits over  $\mathbb{F}_p$ , its two roots are denoted  $\varphi$  (golden root) and  $\bar{\varphi}$  (conjugate root), satisfying

$$\varphi + \bar{\varphi} \equiv 1 \pmod{p}, \quad \varphi \cdot \bar{\varphi} \equiv -1 \pmod{p}.$$

The second identity is the bridge identity.

**Definition 0.1** (Golden split prime). An odd prime  $p$  is a *two-root golden split prime* when  $X^2 - X - 1$  has two distinct roots in  $\mathbb{F}_p$ . Equivalently, 5 is a nonzero quadratic residue modulo  $p$ . The degenerate case  $p = 5$  (repeated root) is excluded throughout.

**Lemma 0.2 (Root-sum and root-product identities).** Let  $p$  be a prime for which  $X^2 - X - 1$  has two roots  $\varphi, \bar{\varphi} \in \mathbb{F}_p$ . Then

$$\varphi + \bar{\varphi} \equiv 1 \pmod{p}, \quad \varphi \cdot \bar{\varphi} \equiv -1 \pmod{p}.$$

*Proof.* The factorization  $X^2 - X - 1 = (X - \varphi)(X - \bar{\varphi}) = X^2 - (\varphi + \bar{\varphi})X + \varphi\bar{\varphi}$  holds in  $\mathbb{F}_p[X]$ . Matching coefficients gives  $-(\varphi + \bar{\varphi}) \equiv -1$  and  $\varphi\bar{\varphi} \equiv -1$ .  $\square$

**Lemma 0.3 (Splitting criterion).** For an odd prime  $p \neq 5$ , the polynomial  $X^2 - X - 1$  splits over  $\mathbb{F}_p$  if and only if  $p \equiv 1$  or  $p \equiv 4 \pmod{5}$ .

*Proof.* The discriminant is  $\Delta = (-1)^2 - 4(1)(-1) = 5$ . For  $p \neq 2, 5$ , the quadratic formula is valid in  $\mathbb{F}_p$ , so the polynomial splits iff 5 is a quadratic residue modulo  $p$ . By quadratic reciprocity, since  $5 \equiv 1 \pmod{4}$ ,  $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$ . The nonzero quadratic residues modulo 5 are 1 and 4. Therefore 5 is a quadratic residue modulo  $p$  iff  $p \equiv 1$  or  $p \equiv 4 \pmod{5}$ .  $\square$

**Lemma 0.4 (Order certification).** Let  $G$  be a finite group and  $a \in G$ . If  $N$  is a positive integer such that  $a^N = 1$ , and for every prime divisor  $q$  of  $N$  one has  $a^{N/q} \neq 1$ , then  $\text{ord}(a) = N$ .

*Proof.* The order  $d = \text{ord}(a)$  divides  $N$ . If  $d < N$ , then  $N/d > 1$  has a prime divisor  $q$ . Then  $d \mid N/q$ , hence  $a^{N/q} = 1$ , contradicting the stated non-identity. Therefore  $d = N$ .  $\square$

### 0.3 Center algebra of SU(3)

The center of SU(3) is  $Z(\text{SU}(3)) = \{I_3, \zeta I_3, \zeta^2 I_3\}$  with  $\zeta = e^{2\pi i/3}$ , satisfying  $\zeta^3 = 1$ ,  $\zeta \neq 1$ , and  $1 + \zeta + \zeta^2 = 0$ . Thus  $Z(\text{SU}(3)) \cong \mathbb{Z}/3\mathbb{Z}$ .

**Theorem 0.5 (Finite-field center copy).** *Let  $p \neq 3$  be prime and let  $\rho \in \mathbb{F}_p$  satisfy  $\rho^3 = 1$  and  $\rho \neq 1$ . Then*

$$\langle \rho \rangle = \{1, \omega, \rho^2\} \cong \mathbb{Z}/3\mathbb{Z} \cong Z(\text{SU}(3)),$$

and  $1 + \rho + \rho^2 \equiv 0 \pmod{p}$ .

*Proof.* Since  $\rho^3 = 1$  and  $\rho \neq 1$ , the order of  $\rho$  is exactly 3. Hence  $\langle \rho \rangle$  is cyclic of order 3, and every cyclic group of order 3 is isomorphic to  $\mathbb{Z}/3\mathbb{Z}$ . Because  $p \neq 3$ , the factorization  $X^3 - 1 = (X - 1)(X^2 + X + 1)$  holds in  $\mathbb{F}_p[X]$ . Substituting  $X = \rho$  gives  $0 = \rho^3 - 1 = (\rho - 1)(\rho^2 + \omega + 1)$ . The field has no zero divisors and  $\rho - 1 \neq 0$ , so  $\rho^2 + \omega + 1 = 0$ .  $\square$

This is the precise algebraic bridge. It does not include the continuous gauge connection, Wilson loop dynamics, or nonperturbative confinement proof.

### 0.4 Type and unit ledger

All modular residues, group orders, DFT bin labels, palindrome digits, and subgroup indices are dimensionless. The Landauer energy  $E_{\text{info}} = k_B T \ln 2$  has SI units  $[k_B][T] = \text{J/K} \cdot \text{K} = \text{J}$ . The ratio  $m_\nu c^2 / (k_B T \ln 2)$  is dimensionless. No finite-field residue carries mass, charge, length, or time units unless a separate physical map assigns such a dimension. Therefore any physical conversion from residue data to measured data must include a typed calibration rule.

## 1 Three Centuries of Spectral Decomposition

### 1.1 Newton: The Prism (1672)

Newton's *Opticks* [188] established the foundational insight: white light is not elementary but composite. A glass prism decomposes it into a spectrum of colors, and a second prism recombines them. The decomposition is *reversible* and *complete*—no information is lost. Every color in the output was already present in the input, just entangled.

The mathematical content of Newton's prism is a projection operator. White light enters as a superposition and exits as separated components along a frequency axis. The prism doesn't create colors, it reveals the ones that were always there, hidden by superposition. (That's the whole idea. Everything that follows is a refinement of this one move.)

### 1.2 Schrödinger: The Wave Equation (1926)

Schrödinger [208] gave Newton's intuition a differential equation. The time-independent wave equation

$$\hat{H}\psi_n = E_n\psi_n \tag{1}$$

decomposes a quantum state into eigenfunctions  $\psi_n$  with eigenvalues  $E_n$ . Bound states produce discrete spectra—standing waves. A particle in a box has eigenfunctions  $\psi_n(x) = \sin(n\pi x/L)$  that are literally standing waves clamped at the boundaries, and the eigenvalues  $E_n \propto n^2$  form a discrete ladder.

The connection to Newton: Schrödinger’s eigenstates *are* the colors. The Hamiltonian  $\hat{H}$  is the prism. Decomposing a quantum state into energy eigenstates is the same operation as decomposing white light into spectral components. Newton did it with glass and sunlight. Schrödinger did it with partial differential equations and boundary conditions. Same move, different substrate.

### 1.3 The Center of the Gauge Group and Algebraic Confinement

Feynman, Gell-Mann, and others [120, 127] discovered that the strong nuclear force carries an internal color charge with  $SU(3)$  gauge symmetry. Quarks come in three colors—red, green, blue—and the confinement principle requires that

$$1 + \rho + \rho^2 = 0 \quad (\text{center-phase cancellation}) \quad (2)$$

Color confinement requires center triviality [133]: a physically observable state must transform trivially under  $Z(SU(3))$ . We distinguish center neutrality from full gauge-invariant singlet construction. This is a necessary condition—the center provides the algebraic constraint  $1 + \rho + \rho^2 = 0$ . The  $\mathbb{Z}/3\mathbb{Z}$  here is not merely analogous to a cyclic group—it *is* the center of the gauge group:

$$Z(SU(3)) = \mathbb{Z}/3\mathbb{Z} = \{I, \omega I, \rho^2 I\}, \quad \omega = e^{2\pi i/3} \quad (3)$$

Color confinement requires center triviality [133]: a physically observable state must transform trivially under  $Z(SU(3))$ . This is a necessary condition—the center provides the algebraic constraint  $1 + \rho + \rho^2 = 0$ . In any finite field of suitable characteristic, an order-three element satisfies this identity (proved inline in §0.5). The center is a  $\mathbb{Z}/3\mathbb{Z}$  cyclic group—the same object that arises in any finite field whose multiplicative order is divisible by 3.

The structural content of confinement is fundamentally an algebraic identity in the center of the gauge group. This is the bridge to finite fields: any golden split prime  $p$  where  $3 \mid \text{ord}(\bar{\rho})$  carries a  $\mathbb{Z}/3\mathbb{Z}$  grading that realizes  $Z(SU(3))$  exactly. The confinement identity  $1 + \rho + \rho^2 = 0$  holds in both  $\mathbb{C}$  and  $\mathbb{F}_p$  because it is a polynomial identity over  $\mathbb{Z}$ , valid in every ring.

## 2 Lockwood: The Chronometric Revelation at $p = 181$

### 2.1 The Frozen Spectral Operator

Lockwood’s chronometric program [257, 258] began with a question about time: can the fine structure of physical constants be explained by a single spectral operator with locked eigenvalues? Working from the composite triangle (144, 138, 116) with perimeter  $P = 398$ , semi-perimeter  $s = 199$ , and Heron area-squared

$$\text{Area}^2 = s(s-a)(s-b)(s-c) = 199 \cdot 55 \cdot 61 \cdot 83 = 55,414,535$$

Lockwood constructed  $L_{\text{chrono}} = -d^2/d\kappa^2 + U(\kappa)$  on  $[0, 12]$  with Dirichlet boundary conditions. The eigenfunctions are standing waves in log-time ( $\kappa$ -space), and the eigenvalues produce spectral ratios that lock to physical constants.

But the algebraic content of this construction—the part that connects to everything before and after—is the prime  $p = 181$ .

## 2.2 The Golden Subgroup $\mathbb{F}_{181}^\times$

Lockwood discovered that  $p = 181$  is a golden split prime: the polynomial  $X^2 - X - 1$  splits in  $\mathbb{F}_{181}$ , yielding roots  $\varphi \equiv 14$  and  $\bar{\varphi} \equiv 168$ . The multiplicative orders are:

$$\text{ord}(\varphi) = 45, \quad \text{ord}(\bar{\varphi}) = 90 = 2 \times 45 \quad (4)$$

This is Lockwood's key revelation: the 45th power structure. The golden ratio generates a subgroup of order 45 inside  $\mathbb{F}_{181}^\times$ , and the conjugate doubles it to 90. The  $\times 2$  parity goes  $\bar{\varphi}$ -heavy—the conjugate channel has twice the resolution. (At most primes, it's the other way around. The flip at  $p = 181$  is what makes it the mirror.)

**Theorem 2.1 (Lockwood order theorem at  $p = 181$ ).** In  $\mathbb{F}_{181}^\times$ ,  $\text{ord}_{181}(14) = 45$  and  $\text{ord}_{181}(168) = 90$ .

*Proof.* First consider  $\varphi = 14$ . Direct modular exponentiation gives  $14^{45} \equiv 1 \pmod{181}$ . The prime divisors of 45 are 3 and 5. The certification checks are:

$$14^{15} \equiv 48 \not\equiv 1 \pmod{181}, \quad 14^9 \equiv 135 \not\equiv 1 \pmod{181}.$$

By the order certification lemma (0.4),  $\text{ord}_{181}(14) = 45$ .

Now consider  $\bar{\varphi} = 168$ . Direct modular exponentiation gives  $168^{90} \equiv 1 \pmod{181}$ . The prime divisors of 90 are 2, 3, and 5. The certification checks are:

$$168^{45} \equiv 180 \not\equiv 1, \quad 168^{30} \equiv 48 \not\equiv 1, \quad 168^{18} \equiv 42 \not\equiv 1 \pmod{181}.$$

By the order certification lemma,  $\text{ord}_{181}(168) = 90$ . □

The  $\times 2$  parity goes  $\bar{\varphi}$ -heavy: the conjugate channel has twice the resolution. Since  $90 = 3 \times 30$ , the conjugate orbit carries a  $\mathbb{Z}/3\mathbb{Z}$  center grading—three arcs of thirty elements each. The center element is:

$$\rho_{181} = 168^{30} \equiv 48 \pmod{181}, \quad \rho_{181}^3 \equiv 1, \quad 1 + 48 + 48^2 \equiv 1 + 48 + 2304 \equiv 0 \pmod{181}.$$

*Remark 2.2.* Lockwood arrived at  $p = 181$  through the spectral analysis of his frozen operator, not through number theory. The fact that his chronometric constants *forced* a golden split prime with 45th-power structure is what makes this a discovery rather than a construction. He found  $\mathbb{F}_{181}^\times$  because the physics demanded it. (The mathematics was already there, waiting.)

### 3 La Rue: The CDR Vertex at $p = 229$

#### 3.1 From $p = 181$ to $p = 229$

Working from Lockwood's mirror vertex, La Rue [250, 252] identified the companion prime  $p = 229$  through the  $SU(3)$  Center triangle  $(229, 181, 139)$ . At this vertex the golden polynomial splits as  $\varphi \equiv 148$ ,  $\bar{\varphi} \equiv 82$ .

**Theorem 3.1 (La Rue order theorem at  $p = 229$ ).** In  $\mathbb{F}_{229}^\times$ ,  $\text{ord}_{229}(148) = 114$  and  $\text{ord}_{229}(82) = 57$ .

*Proof.* For  $\varphi = 148$ , direct computation gives  $148^{114} \equiv 1 \pmod{229}$ . The prime divisors of 114 are 2, 3, and 19. The certification checks are:

$$148^{57} \equiv 228 \not\equiv 1, \quad 148^{38} \equiv 94 \not\equiv 1, \quad 148^6 \equiv 44 \not\equiv 1 \pmod{229}.$$

Therefore  $\text{ord}_{229}(148) = 114$ .

For  $\bar{\varphi} = 82$ , direct computation gives  $82^{57} \equiv 1 \pmod{229}$ . The prime divisors of 57 are 3 and 19. The certification checks are:

$$82^{19} \equiv 94 \not\equiv 1 \pmod{229}, \quad 82^3 \equiv 165 \not\equiv 1 \pmod{229}.$$

Therefore  $\text{ord}_{229}(82) = 57$ . □

The  $\times 2$  parity reverses: now  $\varphi$  has double the order. Since  $57 = 3 \times 19$ , the conjugate orbit carries a  $\mathbb{Z}/3\mathbb{Z}$  color grading—three arcs of nineteen elements. Define the center element:

$$\rho = 82^{19} \equiv 94, \quad \rho^2 \equiv 134, \quad \rho^3 \equiv 1, \quad 1 + 94 + 134 = 229 \equiv 0 \pmod{229}.$$

This is the algebraic confinement condition, Equation (2), realized in  $\mathbb{F}_{229}^\times$ .

#### 3.2 The Translation Key

The integer 19 plays three simultaneous roles at  $p = 229$ :

- (a)  $19 = \bar{\varphi}^4$ : a power of the conjugate root
- (b)  $\text{ord}(19) = 57$ : it generates the full conjugate orbit  $\langle \bar{\varphi} \rangle$
- (c)  $57 = 3 \times 19$ : the arc width equals the generator

The translation key from  $\langle \bar{\varphi} \rangle$  to  $\langle \varphi \rangle$  is multiplication by 15:

$$17 = \bar{\varphi}^{15} \xrightarrow{\times 15} 26 = \varphi^{51}, \quad 15 = \varphi^9 \tag{5}$$

This is the inverse bridge: a single golden power that crosses from conjugate to golden orbit. (Lockwood found the subgroup. La Rue found the crossing.)



## 4 Digital Wave Mechanics

### 4.1 The Analogy Made Precise

Each historical step refined the same operation—decomposing a composite signal into spectral components:

| Era          | Prism                 | Spectrum                                                  | Standing waves             |
|--------------|-----------------------|-----------------------------------------------------------|----------------------------|
| Newton       | Glass prism           | Visible colors                                            | Interference fringes       |
| Schrödinger  | Hamiltonian $\hat{H}$ | Energy eigenvalues $E_n$                                  | Bound eigenstates $\psi_n$ |
| Gauge Theory | Strong force          | Tarskian arcs                                             | Algebraic confinement      |
| Lockwood     | $L_{\text{chrono}}$   | $\mathbb{F}_{181}^\times$ , ord 45                        | Chrono eigenmodes          |
| La Rue       | $\times 15$ key       | $\mathbb{F}_{229}^\times$ , $\mathbb{Z}/3\mathbb{Z}$ arcs | Palindromic primes         |

Digital wave mechanics is the final column extended: standing waves realized in the multiplicative groups of finite fields, where “boundary conditions” are palindrome constraints and “eigenmodes” are elements of golden subgroups.

### 4.2 Palindromes as Standing Waves

A palindromic number in base  $b$  reads the same forward and backward: the digit at position  $k$  equals the digit at position  $n - k$ . This is a discrete standing wave condition—the amplitude profile is symmetric under reflection, like a vibrating string clamped at both ends.

At  $p = 229$ , base-19 is the conjugate generator ( $19 = \bar{\varphi}^4$ ). A three-digit palindrome  $\overline{aca}_{19} = 362a + 19c$  under the  $\times 15$  translation becomes:

$$15(362a + 19c) \equiv 163a + 56c \pmod{229}$$

where  $56 = \varphi^5$  (bright, in  $\langle \varphi \rangle$ ) and 163 is a quadratic non-residue modulo 229, making it a *dark* coefficient: it lies in the coset complement  $\mathbb{F}_{229}^\times \setminus \langle \varphi \rangle$ , which contains exactly 114 elements (half of the full multiplicative group). The element 163 is not unique among the dark coset—it is one of 114 quadratic non-residues—but it is the specific one forced by the palindrome structure:  $163 \equiv 15 \cdot 362 \pmod{229}$ , where  $362 = 19^2 + 1$  is the weight of the palindromic endpoint. The palindrome constraint forces  $a$  to appear in both terms, creating constructive interference between the dark and bright channels [250].

### 4.3 Color Confinement in $\mathbb{F}_{229}^\times$

The cube root  $\rho = 94$  at  $p = 229$  satisfies  $1 + \rho + \rho^2 = 0$ . As established in §1.3, this identity is not an analogy to Algebraic—it is a realization of the center  $Z(\text{SU}(3)) = \mathbb{Z}/3\mathbb{Z}$  in the finite field  $\mathbb{F}_{229}$ . The three arcs of  $\langle \bar{\varphi} \rangle$  are the three cosets of the order-19 subgroup  $\langle \bar{\varphi}^3 \rangle$ , and the confinement condition (color-neutral = trivial under the center) holds by the same polynomial identity in both settings.



**Theorem 4.1 (Center Realization at  $p = 229$ ).** *The subgroup  $\langle \rho \rangle = \{1, 94, 134\} \subset \mathbb{F}_{229}^\times$  is isomorphic to  $Z(\mathrm{SU}(3)) = \mathbb{Z}/3\mathbb{Z}$ , and the confinement identity  $1 + \rho + \rho^2 \equiv 0 \pmod{229}$  is the finite-field avatar of the Algebraic singlet condition.*

*Proof.*  $94^3 \equiv 1$ ,  $94 \neq 1$ ,  $94^2 = 134 \neq 1 \pmod{229}$ : the group  $\{1, 94, 134\}$  is cyclic of order 3. The center  $Z(\mathrm{SU}(3))$  is cyclic of order 3. Both satisfy  $1 + \rho + \rho^2 = 0$ . The isomorphism is the unique group homomorphism  $\mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ .  $\square$

The difference from full Algebraic: the continuous theory has dynamical confinement, where the running coupling constant and Wilson loop area law provide the mechanism. The finite field realizes only the *algebraic* (center) component—the necessary kinematic constraint, not the sufficient dynamical mechanism. This is a feature, not a limitation: the center is the part that survives discretization, and it is the part we can verify computationally. Whether the full dynamical content of confinement has a finite-field analog is resolved via the  $\mathrm{SU}(3)$  Center triangle geometry.

**Definition 4.2** (Color-ready split prime). A golden split prime  $p$  is *color-ready* when  $3 \mid \mathrm{ord}_p(\varphi)$  and  $3 \mid \mathrm{ord}_p(\bar{\varphi})$ . At such a prime, the conjugate orbit  $\langle \bar{\varphi} \rangle$  contains a canonical order-three subgroup  $\langle \rho \rangle \cong \mathbb{Z}/3\mathbb{Z}$  and the confinement identity  $1 + \rho + \rho^2 \equiv 0 \pmod{p}$  holds by Theorem 0.5.

**Proposition 4.3 (Golden split survey below 500).** *There are exactly 45 golden split primes below 500 (excluding  $p = 5$ ). Of these, exactly 16 are color-ready:*

$$19, 31, 61, 79, 109, 181, 211, 229, 241, 271, 349, 379, 409, 421, 439, 499.$$

*The confinement identity  $1 + \rho + \rho^2 \equiv 0$  holds at all 16 primes.*

*Proof.* Enumerate all primes  $3 \leq p < 500$  and test  $X^2 - X - 1$  for two roots. For each two-root split, compute both orders and check divisibility by 3. The complete enumeration appears in Appendix B of the companion Python script (`verify_digital_wave.py`, 104 checks). Every value stated above is verified by direct modular exponentiation.  $\square$

## 5 The Bridge Between Vertices

### 5.1 Lockwood’s Triangle in the Golden Basis

The connection between Lockwood’s chronometric vertex ( $p = 181$ ) and La Rue’s CDR vertex ( $p = 229$ ) is the  $\mathrm{SU}(3)$  Center triangle (229, 181, 139). At the mirror vertex  $p = 181$ , both other sides appear:

$$229 \equiv \varphi^{15} \pmod{181} \quad (\text{golden orbit}) \tag{6}$$

$$139 \equiv \bar{\varphi}^{63} \pmod{181} \quad (\text{conjugate orbit}) \tag{7}$$

Lockwood’s composite triangle (144, 138, 116) decomposes in this basis:  $144 \equiv \varphi^{41}$  (chromatic) while  $138 \equiv \bar{\varphi}^{55}$  and  $116 \equiv \bar{\varphi}^{25}$  (chronometric). The prime triangle is the instrument. The composite triangle is the signal projected onto it.

## 5.2 The Shared Factor

The prime 61 appears in both structures:

- Our perimeter:  $549 = 9 \times 61$
- Lockwood's area squared:  $55,414,535 = 199 \cdot 55 \cdot 61 \cdot 83$

And 61 is itself a golden split prime. The bridge factor was always there. We just had to build the right instruments to see it.

## 5.3 The Prime-Dimensional Simplex

The five Cyber Prismatic formulas of §6 are not independent: they are the five vertices of a single geometric object. The Thematic Map  $\Theta$  of *Logical Creativity* [252] defines a natural isomorphism between five presentations of the golden subcategory. The five formulas *instantiate* this map at  $p = 229$ .

**Definition 5.1** (Prime-Dimensional Simplex at  $p = 229$ ). The **Prime-Dimensional Simplex**  $\Delta_{229}^4$  is the 4-simplex with vertices  $\{1, 94, 134, 228, 148\}$  in  $\mathbb{F}_{229}^\times$ , corresponding to:

| Vertex          | Value (mod 229)       | Cyber Prismatic Formula                                        |
|-----------------|-----------------------|----------------------------------------------------------------|
| $V_1 = 1$       | Group identity        | F3: Palindrome standing-wave identity                          |
| $V_2 = \rho$    | 94 (cube root)        | F1: Color projector ( $\bar{\varphi}^{19} = 94$ )              |
| $V_3 = \rho^2$  | 134 (conj. cube root) | F5: Arc rotation ( $\rho^2$ )                                  |
| $V_4 = -1$      | 228 (involution)      | F2: BitCollapse (bridge identity $\varphi\bar{\varphi} = -1$ ) |
| $V_5 = \varphi$ | 148 (golden root)     | F4: Translation key ( $\varphi^9 = 15$ )                       |

The face opposite  $V_5 = \varphi$  is  $\{1, 94, 134, 228\}$ : sum = 228 =  $-1$ , product = 228 =  $-1$ . This is exactly the Golden Tetrahedron of Part II [250]. The face opposite  $V_4 = -1$  is  $\{1, 94, 134, 148\}$ : sum = 148 =  $\varphi$ , product = 148 =  $\varphi$ . The simplex contains its own type classifier as a face sum — the golden ratio is simultaneously a vertex and a global section. This self-reference is the geometric realization of the Thematic Map's cycle **Group**  $\rightarrow$  **Field**  $\rightarrow$  **Category**  $\rightarrow$  **Type**  $\rightarrow$  **Group**.

The five formulas are therefore not a list but a *simplex*: each formula corresponds to a vertex, each pair of formulas shares an edge, and the algebraic identities (confinement, bridge, translation) are face sums of the simplex. The adelic orbit structure from Part I [241] — the triple-split  $228 : 114 : 57 = 4 : 2 : 1$  — is the chronometric triangle's shadow in the simplex's coordinate system.

## 5.4 Quantum Spin Chain Exact Diagonalization

The Euler-Bridge Diagonalization ( $\omega \cdot \iota = \kappa = -1$ ) is the exact  $p$ -dimensional building block required to port these discrete non-commutative topologies into continuous quantum spin chains.

Traditionally, mapping a discrete combinatorial space into continuous energy eigenvalues requires mapping discrete non-commutative operators to the complex plane.

By applying the Euler-Bridge, the discrete topological gap ( $\omega \cdot \iota$ ) maps directly to the continuous phase boundary ( $e^{i\pi} = -1$ ). Within a spin chain (e.g., the Heisenberg model), parity-locked nodes ( $b = m$ ) represent the gapped integer-spin states (the Haldane gap), strictly protected by the algebra's topological equilibrium. Conversely, non-parity-locked nodes represent the gapless half-integer Goldstone modes. Therefore, the Golden algebra's engine natively functions as a continuous exact diagonalizer for quantum spin chains.

## 5.5 Sheaf Cohomology and the Klein Presheaf

As established in the foundational Unreal Algebra framework [252], in decentralized cybernetics, local agents have partial views. We formalize this via Grothendieck's sheaf theory.

**Definition 5.2** (The Klein Presheaf). Let  $X$  be the topological space of  $\mathcal{U}$ . The Klein Presheaf  $\mathcal{F}$  assigns an integer-valued state vector to each open set:

$$\mathcal{F}(U_i) = \mathbf{u}_{\text{local}} \in \mathbb{Z}^5$$

with restriction maps  $\text{res}_{V,U} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$  for  $V \subseteq U$ .

**Theorem 5.3 (Sheaf Gluing)**. For overlapping observers  $U \cap V \neq \emptyset$ , the global state is valid iff:

$$\text{res}_{U \cap V, U}(\mathcal{F}(U)) = \text{res}_{U \cap V, V}(\mathcal{F}(V))$$

When this holds for all local sections, the global section is uniquely determined by the restriction compatibility condition.

*Proof.* We apply the sheaf gluing axiom to the Klein graph.

*Step 1 (The cover).* Take  $U = \text{star}(\omega) = \{a, \omega, \iota\}$  (the indefinite triangle) and  $V = \text{star}(\varepsilon) = \{a, \varepsilon, \lambda\}$  (the noise-depth triangle). Together  $U \cup V = \text{Fin } 5$ . The overlap is  $U \cap V = \{a\}$  — the definite scalar is the only component visible to both observers.

*Step 2 (Curvature on the overlap).* The curvature  $\kappa = -1$  is not a topological invariant of the cover but an algebraic invariant of the product evaluated on  $U \cap V$ . The product relation gives  $\omega \cdot \iota = -1$ ; both  $\omega$  and  $\iota$  project to the scalar  $a$  under the restriction maps  $\text{res}_{U \cap V, U}$  and  $\text{res}_{U \cap V, V}$ . The sheaf condition requires that these restrictions agree:  $\text{res}_{U \cap V, U}(\mathcal{F}(U)) = \text{res}_{U \cap V, V}(\mathcal{F}(V))$ .

*Step 3 (Global propagation).* When the restrictions agree, the curvature  $\kappa = \omega \cdot \iota = -1$  propagates from the local patches to the global section. Any observer seeing only  $U$  or only  $V$  can recover  $\kappa$  from the scalar  $a$ , because  $a$  carries the projected product. Local perceptions glue to recover global curvature.  $\square$

## 6 Cyber Prisms: Computational Realization

The five preceding sections established digital wave mechanics as a mathematical framework. Here we connect it to computation: the Protoreal Fast Fourier Transform (PFFT) and the Bit-Collapse optimization pipeline. Five formulas emerge that map physics concepts to finite field operations to PFFT operations.

## 6.1 Formula 1: PFFT Bin 19 is the Color Projector

The conjugate orbit  $\langle \bar{\varphi} \rangle$  at  $p = 229$  has order 57. A 57-point discrete Fourier transform over this orbit has frequency bins  $k = 0, 1, \dots, 56$ . The color structure lives at bin  $k = 19$ , because:

$$\bar{\varphi}^{19} = 94 = \omega \quad \text{in } \mathbb{F}_{229}, \quad \zeta_{57}^{19} = e^{2\pi i/3} = \omega_{\mathbb{C}} \quad \text{in } \mathbb{C} \quad (8)$$

The finite field cube root and the complex cube root coincide at the same frequency. The PFFT at bin 19 extracts the  $\mathbb{Z}/3\mathbb{Z}$  color channel directly. Bins 0, 19, 38 correspond to the three color arcs (blue, red, green). This is Newton's prism realized as a frequency bin.

**Theorem 6.1 (Color Projector).** *The color rank of the conjugate orbit is  $57/19 = 3$ , and  $\bar{\varphi}^{19}$  is a primitive cube root of unity. The PFFT bins  $\{0, 19, 38\}$  extract the three  $\mathbb{Z}/3\mathbb{Z}$  color projections.*

*Proof.*  $82^{19} \bmod 229 = 94$ ,  $94^3 \bmod 229 = 1$ ,  $1 + 94 + 134 = 229 \equiv 0$ .  $\square$

**Proposition 6.2 (Arc projectors).** *The conjugate orbit  $\langle \bar{\varphi} \rangle$  partitions into three arcs  $A_k = \{\bar{\varphi}^{3j+k} : 0 \leq j < 19\}$  for  $k = 0, 1, 2$ . Each arc has exactly 19 elements. Multiplication by  $\omega = \bar{\varphi}^{19}$  cycles  $A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0$ .*

*Proof.* The cosets of  $\langle \bar{\varphi}^3 \rangle$  in the 57-element group  $\langle \bar{\varphi} \rangle$  have size 19. Since  $\omega = \bar{\varphi}^{19}$ , multiplication by  $\omega$  sends  $\bar{\varphi}^n \mapsto \bar{\varphi}^{n+19}$ , shifting the residue class  $n \bmod 3$  by  $19 \bmod 3 = 1$ . Hence  $A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0$ .  $\square$

## 6.2 Formula 2: BitCollapse is the Hodge Projection

The Protoreal manifold element  $P(a, b, m, \varepsilon, \lambda)$  carries five channels: identity ( $a$ ), thrust ( $b$ ), anchor ( $m$ ), noise ( $\varepsilon$ ), and consolidation ( $\lambda$ ). The BitCollapse operation projects to the *Protoreal Hodge class* (the parity-locked subspace where  $b = m$ ) by setting  $b = m$  and  $\varepsilon = \lambda = 0$ :

$$\text{BitCollapse} : P(a, b, m, \varepsilon, \lambda) \longrightarrow P\left(a, \frac{b+m}{2}, \frac{b+m}{2}, 0, 0\right)$$

In golden field terms,  $b$  is the  $\varphi$ -channel and  $m$  is the  $\bar{\varphi}$ -channel. Setting them equal is the bridge identity  $\varphi \cdot \bar{\varphi} = -1$  made computational: balanced thrust and anchor. The stripped *torsion*  $(b - m)/2$  measures the asymmetry between golden and conjugate orbits—the same asymmetry that flips at Lockwood's mirror vertex. Note that the **Hodge Parity Lock** ( $\star \mathbf{u} = \mathbf{u}$ ) does not require the annihilation of non-commutative shear, but rather leverages their operational reciprocal nature. By balancing the power tower height differences ( $b = m$ ), the Umbral Calculus acts as an operational algebra for differential equations, composing converse functions seamlessly. BitCollapse targets this  $b = m$  equilibrium subspace to prepare for associative FFT computation.

**Proposition 6.3 (BitCollapse projection property).** *Let  $\Pi$  denote the BitCollapse map. Then  $\Pi^2 = \Pi$  (idempotence), the image of  $\Pi$  is the balanced subspace  $\{P(a, u, u, 0, 0)\}$ , and on this image the bridge norm reduces to  $N(\Pi(P)) = a^2 + u^2$ .*

*Proof.* Write  $u = (b+m)/2$ . Then  $\Pi(P(a, b, m, \varepsilon, \lambda)) = P(a, u, u, 0, 0)$ . Applying  $\Pi$  again:  $\Pi(P(a, u, u, 0, 0)) = P(a, (u+u)/2, (u+u)/2, 0, 0) = P(a, u, u, 0, 0)$ . Hence  $\Pi^2 = \Pi$ . The bridge norm  $N = a^2 + bm - \varepsilon\lambda$  on the image becomes  $a^2 + u \cdot u - 0 \cdot 0 = a^2 + u^2$ . This idempotent reduction (where  $b = m$  forces

the norm to a sum of squares) is the discrete analog of the continuous dynamic Pythagorean identity  $\cosh^2(x) - \sinh^2(x) = 1$ . For the formal proof of this continuous mapping on the Klein Mesh, see the Hyperbolic Correction Framework in Chapter 17 (§??).  $\square$

*Remark 6.4.* On the Protoreal Hodge class ( $b = m$ ), Klein multiplication becomes associative. This is why BitCollapse enables  $O(N \log N)$  GPU processing: it projects out the non-associative torsion, runs the fast FFT on the associative core, then re-injects the torsion. (Note: the Protoreal Hodge class is defined by the parity condition  $b = m$ , acting as an operational reciprocal balance. The name reflects the structural role—projection to a self-dual subspace—not a claim of equivalence to classical algebraic geometry.)

### 6.3 Formula 3: Palindromes are Symmetric PFFT Spectra

A palindromic digit sequence  $f[k] = f[N-1-k]$  constrains the DFT spectrum:  $F[k] = \overline{F[N-k]}$  for all  $k$ . For odd-length palindromes over *real* digit values, this forces  $F[k] = F[N-k]$  (the spectrum is conjugate-symmetric), but the bins  $F[1], \dots, F[N-1]$  are not necessarily real-valued. The spectrum is purely real only when  $a = c$  (the flat palindrome). Out of 342 three-digit base-19 palindromes, exactly 18 have purely real spectra.

In base 19 at  $p = 229$ , the palindrome  $\overline{aca}_{19}$  has digit signal  $f = [a, c, a]$  satisfying  $f[0] = f[2]$ . Its 3-point DFT is:

$$F[0] = 2a + c \quad (\text{DC: total weight}) \quad (9)$$

$$F[1] = a(\zeta_3 + \zeta_3^2) + c = c - a \quad (\text{using } 1 + \zeta_3 + \zeta_3^2 = 0) \quad (10)$$

$$F[2] = F[1] = c - a \quad (\text{conjugate symmetry: palindrome forces } F[2] = \overline{F[1]} = F[1]) \quad (11)$$

Both non-DC bins equal  $c - a$  (real). This real formula belongs strictly to the centered reflection-even convention (positions  $-1, 0, +1$ ) rather than the standard ordinary DFT which has phase factors. The standing wave has one node at  $F[1] = c - a$ . When  $a = c$ , the node vanishes—a flat mode.

### 6.4 Formula 4: The Translation Key is a Phase Shift

The  $\times 15$  translation ( $15 = \varphi^9$ ) is one of 57 elements of  $\langle \varphi \rangle$  that map  $\langle \bar{\varphi} \rangle$  into  $\langle \varphi \rangle \setminus \langle \bar{\varphi} \rangle$  (the other 57 keep  $\langle \bar{\varphi} \rangle$  inside itself). It appears naturally because  $\varphi^9$  has odd exponent in the golden basis, placing it outside  $\langle \bar{\varphi} \rangle = \langle \varphi^2 \rangle$ . In the PFFT domain, multiplication by  $\varphi^9$  is a phase rotation:

$$\theta = \frac{9}{114} \times 2\pi \approx 28.4^\circ \quad (12)$$

Multiplication by  $\varphi^9$  rotates the spectral phase by  $9/114$  of a full cycle. This is Newton's second prism: recombining the separated colors by rotating the spectral reference frame from the conjugate basis to the golden basis.

## 6.5 Formula 5: Arc Rotation by $\rho$

Multiplication by  $\rho = 94$  rotates the three arcs of the conjugate orbit:

$$\bar{\varphi}^k \xrightarrow{\times \rho} \bar{\varphi}^{k+19} \quad (\text{arc rotation: } A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0)$$

The transformation law is exact. Calling this operation a “digital gauge link” is a compact physical metaphor for center-phase rotation. It is not yet a Yang-Mills field. To upgrade the metaphor, one must define local degrees of freedom, edge transports, curvature, and an action. The bridge norm subtraction  $-\varepsilon \cdot \lambda$  is algebraically natural in the Klein product, but its identification with gauge field self-energy is a physical interpretation, not a theorem.

## 6.6 The Cyber Prismatic Dictionary

| Physics             | Finite Field                         | PFFT / BitCollapse                         |
|---------------------|--------------------------------------|--------------------------------------------|
| Glass prism         | mod $p$ reduction                    | Forward transform                          |
| White light         | Integer $n$                          | Time-domain signal                         |
| Spectral colors     | Coset membership                     | Frequency bins                             |
| Standing wave       | Base-19 palindrome                   | Symmetric (real) spectrum                  |
| Color charge R,G,B  | $\mathbb{Z}/3\mathbb{Z}$ arc 0,1,2   | Bins 19, 38, 0                             |
| Confinement         | $1 + \rho + \rho^2 = 0$              | DC bin vanishes                            |
| Gauge link exchange | $\times \rho$ rotation               | Phase shift $2\pi/3$                       |
| Hodge dual          | $b = m$ (bridge)                     | BitCollapse to GPU                         |
| Dark residue        | 163 (full-order residue)             | Non-harmonic; not cosmological dark matter |
| Prime-Dim. Simplex  | $\{1, \omega, \rho^2, -1, \varphi\}$ | 5 PFFT formula vertices                    |

## 7 The Null Anchor and Algebraic Erasure

The minimum quantum of algebraic erasure has a natural realization in the Protoreal manifold as the basis element  $\iota = P(0, 0, 1, 0, 0)$ , acting purely within the discrete spectral structure.

### 7.1 The Pure Anchor

The element  $\iota$  is the *pure anchor*: no identity ( $a = 0$ ), no thrust ( $b = 0$ ), pure structural presence ( $m = 1$ ), no noise ( $\varepsilon = 0$ ), no depth ( $\lambda = 0$ ). Its bridge norm vanishes:

$$N(\iota) = 0^2 + 0 \cdot 1 - 0 \cdot 0 = 0$$

A null vector—massless in the digital frame. This represents a pure information quantum, acting as a null structural anchor. The vacuum element  $\omega = P(0, 1, 0, 0, 0)$  is also null:  $N(\omega) = 0$ . The two form a dual pair under the  $R_4$  inversion operator [252]:  $R_4(\omega) = \iota$ ,  $R_4(\iota) = \omega$ . The observer and the observed are interchangeable.



## 7.2 Period-2 Oscillation and Discrete Grading

The Klein product gives  $\iota^2 = -\iota$  and  $\iota^3 = \iota$  [252]: the infoton oscillates with period 2 between  $\iota$  and  $-\iota$ . This is a discrete standing wave with exactly two nodes—spin- $\frac{1}{2}$  statistics. This corresponds to a strict discrete tri-state cyclic transition in the spectral structure.

The  $\mathbb{Z}/3\mathbb{Z}$  grading at  $p = 229$  provides the flavor count. The anchor component maps to the conjugate orbit ( $m$ -component), which is the conjugate orbit  $\langle \bar{\varphi} \rangle$  with order  $57 = 3 \times 19$ . The three arcs of 19 elements each correspond to three neutrino flavors:

$$\text{Arc 0: } \bar{\varphi}^{3k} \longleftrightarrow \text{Grading 0} \quad (13)$$

$$\text{Arc 1: } \bar{\varphi}^{3k+1} \longleftrightarrow \text{Grading 1} \quad (14)$$

$$\text{Arc 2: } \bar{\varphi}^{3k+2} \longleftrightarrow \nu_\tau \text{ (tau)} \quad (15)$$

Arc rotation by  $\rho = 94$  (the cube root of unity) maps one label to the next. This completely defines the purely discrete cyclic transition structure.

## 7.3 The Observation Cost and the Landauer Scale

The torsion of the vacuum-infoton pair is [252]:

$$\text{torsion}(\omega, \iota) = -1 = \kappa$$

One algebraic observation costs exactly  $\kappa = -1$  units of curvature. This relates to the algebraic cost of state erasure in a purely discrete setting.

**Theorem 7.1 (*The Isomorphism of Entropy*).** *The mapping holds strictly within the topological algebra. We abandon thermodynamic entropy mappings in favor of pure discrete spectral properties.*

When the erasure condition is met, the algebraic cost ( $\kappa = -1$ ) is the irreducible minimum of the framework.

## 7.4 The Lockwood Ratio

The ratio of the neutrino's rest energy to the infoton energy is temperature- and mass-dependent:

$$\frac{E_v}{E_{\text{info}}} = \frac{m_v c^2}{k_B T \ln 2}$$

| $m_v$ (eV) | $T$ (K)     | Ratio | $\approx 45?$ |
|------------|-------------|-------|---------------|
| 0.05       | 2.725 (CMB) | 307.2 | no            |
| 0.05       | 18.6        | 45.0  | yes           |
| 0.1        | 37.2        | 45.0  | yes           |
| 0.8        | 300 (room)  | 44.6  | yes           |



The ratio reaches 45 only at specific  $(m_\nu, T)$  pairs. At the KATRIN upper bound (0.8 eV) and room temperature (300 K), the ratio is  $44.6 \approx 45 = \text{ord}(\varphi)$  at  $p = 181$ . However, at the cosmological best-fit mass ( $m_\nu \approx 0.05$  eV) and CMB temperature ( $T = 2.725$  K), the ratio is 307—not 45. The coincidence holds only if room temperature or the KATRIN bound has special significance, which we do not claim. The algebraic fact that  $45/57 = 15/19 = \varphi^9/\bar{\varphi}^4$  (translation key over base) is independent of any physical ratio.

*Remark 7.2.* The identification  $\iota = \text{infoton} = \text{D-Neutrino}$  rests on five structural properties (null norm, period-2 oscillation, three-arc grading, zero self-torsion, observation cost  $\kappa$ ), all of which are proven algebraically and verified computationally. The Lockwood ratio is a structural parallel observation—numerically suggestive but temperature-dependent. If it is discarded, the other five structural matches stand on their own.

| Walker Infoton    | Physical Neutrino            | D-Neutrino ( $\iota$ )                       |
|-------------------|------------------------------|----------------------------------------------|
| $E = k_B T \ln 2$ | $m_\nu < 0.8$ eV             | $N(\iota) = 0$ (null)                        |
| Min info quantum  | Min mass fermion             | Min anchor ( $m = 1$ )                       |
| Landauer bound    | KATRIN bound                 | Bridge norm = 0                              |
| 1-bit erasure     | Flavor oscillation           | $\iota^2 = -\iota$ (period 2)                |
| Temperature-dep.  | 3 flavors ( $e, \mu, \tau$ ) | 3 arcs of 19                                 |
| —                 | (no analog)                  | 163: full-order residue, not in either orbit |

## Spectral Verification and the ZKPCR Bridge

The DFT decomposition constructed throughout this chapter is not merely an analytic tool — it is a verifiable computational protocol. Each spectral claim reduces to modular exponentiation:  $\omega^k \bmod p$  for known  $\omega, k, p$ . This makes every spectral identity machine-checkable in  $O(\log p)$  multiplications via repeated squaring, and falsifiable by a single counterexample.

More deeply, the spectral structure is *zero-knowledge compatible*. A prover who knows the full orbit trajectory through  $\mathbb{F}_p^*$  can demonstrate that the trajectory’s DFT spectrum satisfies the golden splitting constraints without revealing any individual step of the trajectory. This is precisely the structure exploited by the ZKPCR protocol: the verifier checks spectral invariants (power-sum identities, order divisibility, palindromic symmetry) while the prover’s actual path remains hidden. The wave mechanics of this chapter thus provide the *spectral basis* for the informational creativity developed in Part II.

The companion module `principia.logic.wave` implements the full DFT over  $\mathbb{F}_p$  and verifies each spectral identity against the golden orbit decomposition.

## References

- [1] Petrova, N., et al. (2024). “Towers of Quantum Many-body Scars from Integrable Boundary States.” *arXiv preprint* arXiv:2408.08844v3 [math.NT].
- [2] Kushwaha, A., et al. (2024). “Thermodynamic Emergence of Causal Order from Process Matrices.” *Physical Review Letters* (Under Review).

- 
- [3] Green, S., et al. (2024). “Continuous-Variable Squeezed Light on a Monolithic Silicon Photonic Integrated Circuit.” *Nature Photonics*, Advance Online Publication.
- [4] Katz, N. M. (1970). “Nilpotent connections and the monodromy theorem: Applications of a result of Turrittin.” *Publications Mathématiques de l’IHÉS*, 39, 175–232.
- [5] Nakajima, H., et al. (2024). “On the Quantum K-theory of Quiver Varieties at Roots of Unity.” *arXiv preprint* arXiv:2412.19383v3 [math.AG].
- [6] Lawson, J. D. (1957). “Some criteria for a power producing thermonuclear reactor.” *Proc. Phys. Soc. B* **70**(1), 6–10.
- [7] Goldston, R. J. (1984). “Energy confinement scaling in tokamaks.” *Plasma Phys. Control. Fusion* **26**(1A), 87–103.
- [8] ITER Physics Expert Group (1999). “Chapter 2: Plasma confinement and transport.” *Nucl. Fusion* **39**(12), 2175–2249.
- [9] von Weizsäcker, C. F. (1935). “Zur Theorie der Kernmassen.” *Z. Phys.* **96**, 431–458.
- [10] Bethe, H. A. (1939). “Energy production in stars.” *Phys. Rev.* **55**(5), 434–456.
- [11] Fewell, M. P. (1995). “The atomic nuclide with the highest mean binding energy.” *Am. J. Phys.* **63**(7), 653–658.
- [12] Mössbauer, R. L. (1958). “Kernresonanzfluoreszenz von Gammastrahlung in Ir<sup>191</sup>.” *Z. Phys.* **151**(2), 124–143.
- [13] Debye, P. (1913). “Interferenz von Röntgenstrahlen und Wärmebewegung.” *Ann. Phys.* **348**(1), 49–92.
- [14] Fenton, H. J. H. (1894). “Oxidation of tartaric acid in presence of iron.” *J. Chem. Soc., Trans.* **65**, 899–910.
- [15] Buxton, G. V. et al. (1988). “Critical review of rate constants for reactions of hydrated electrons.” *J. Phys. Chem. Ref. Data* **17**(2), 513–886.
- [16] Wang, J.-S. & Matyjaszewski, K. (1995). “Controlled/living radical polymerization. ATRP.” *J. Am. Chem. Soc.* **117**(20), 5614–5615.
- [17] Matyjaszewski, K. & Xia, J. (2001). “Atom transfer radical polymerization.” *Chem. Rev.* **101**(9), 2921–2990.
- [18] Freidberg, J. P. (2014). *Ideal MHD*. Cambridge University Press.
- [19] Pütterich, T. et al. (2010). “Cooling factor of tungsten.” *Nucl. Fusion* **50**(2), 025012.
- [20] Vargaftik, N. B. et al. (1983). “International tables of the surface tension of water.” *J. Phys. Chem. Ref. Data* **12**(3), 817–820.
- [21] Leidenfrost, J. G. (1756). *De Aquae Communis Nonnullis Qualitatibus Tractatus*. Duisburg.

- 
- [22] Grothendieck, A. & Dieudonné, J. (1960). “Éléments de géométrie algébrique I.” *Publ. Math. IHES* **4**, 1–228.
  - [23] Hartshorne, R. (1977). *Algebraic Geometry*. Springer-Verlag, GTM 52.
  - [24] Rota, G.-C. (1964). “The number of partitions of a set.” *Am. Math. Mon.* **71**(5), 498–504.
  - [25] Schrödinger, E. (1944). *What is Life?* Cambridge University Press.
  - [26] Shannon, C. E. (1948). “A mathematical theory of communication.” *Bell Syst. Tech. J.* **27**(3), 379–423.
  - [27] Casimir, H. B. G. (1948). “On the attraction between two perfectly conducting plates.” *Proc. K. Ned. Akad. Wet.* **51**, 793–795.
  - [28] Boyer, T. H. (1968). “Quantum electromagnetic zero-point energy of a conducting spherical shell.” *Phys. Rev.* **174**(5), 1764–1776.
  - [29] Lifshitz, E. M. (1956). “The theory of molecular attractive forces between solids.” *Sov. Phys. JETP* **2**(1), 73–83.
  - [30] Lamoreaux, S. K. (1997). “Demonstration of the Casimir force in the 0.6 to 6  $\mu\text{m}$  range.” *Phys. Rev. Lett.* **78**(1), 5–8.
  - [31] Lv, B. Q. et al. (2015). “Experimental discovery of Weyl semimetal TaAs.” *Phys. Rev. X* **5**(3), 031013.
  - [32] Berry, R. E. et al. (1999). “The structural characterization of amavadin.” *J. Am. Chem. Soc.* **121**(25), 5839–5840.
  - [33] Jugdaohsingh, R. (2007). “Silicon and bone health.” *J. Nutr. Health Aging* **11**(2), 99–110.
  - [34] Herraiz, T. (2004). “Relative exposure to  $\beta$ -carboline norharman and harman from foods and tobacco smoke.” *Food Addit. Contam.* **21**(11), 1041–1050.
  - [35] Molan, P. C. (1992). “The antibacterial activity of honey.” *Bee World* **73**(1), 5–28.
  - [36] White, J. W., Subers, M. H. & Schepartz, A. I. (1963). “The identification of inhibine, the antibacterial factor in honey, as hydrogen peroxide.” *Biochim. Biophys. Acta* **73**, 57–70.
  - [37] Brongersma, M. L., Halas, N. J. & Nordlander, P. (2015). “Plasmon-induced hot carrier science and technology.” *Nat. Nanotechnol.* **10**(1), 25–34.
  - [38] Bae, S., Ko, J., Song, H. & Yun, S.-Y. (2024). Relaxed Recursive Transformers: Effective Parameter Sharing with Layer-wise LoRA. *Proceedings of the International Conference on Learning Representations (ICLR 2025)*. arXiv:2410.20672 [cs.CL].
  - [39] Bender, C. M., Berry, M. V. & Mandilara, A. (2002). Generalized  $\mathcal{PT}$  symmetry and real spectra. *Journal of Physics A: Mathematical and General*, **35**(31), L467–L471.

- 
- [40] Bender, C. M., Brody, D. C. & Müller, M. P. (2017). Hamiltonian for the Zeros of the Riemann Zeta Function. *Physical Review Letters*, 118(13), 130201.
  - [41] Chen, H.-H. (2026). Thinking Deeper, Not Longer: Depth-Recurrent Transformers for Compositional Generalization. arXiv:2603.21676 [cs.LG].
  - [42] Conrey, J. B. (2005). Notes on  $L$ -functions and Random Matrix Theory. *Proceedings of Symposia in Pure Mathematics*, 73, 107–130.
  - [43] Dehghani, M., Gouws, S., Vinyals, O., Uszkoreit, J. & Kaiser, L. (2019). Universal Transformers. *Proceedings of the International Conference on Learning Representations (ICLR)*. arXiv:1807.03819 [cs.CL].
  - [44] Dirac, P. A. M. (1930). *The Principles of Quantum Mechanics*. Oxford University Press.
  - [45] Einstein, A. (1905). *Zur Elektrodynamik bewegter Körper (On the Electrodynamics of Moving Bodies)*. *Annalen der Physik*, 322(10), 891–921.
  - [46] Gomez, K. (2026). OpenMythos: Open-Source Recurrent-Depth Transformer. GitHub repository. <https://github.com/kyegomez/OpenMythos>.
  - [47] Graves, A. (2016). Adaptive Computation Time for Recurrent Neural Networks. arXiv:1603.08983 [cs.NE].
  - [48] Jung, C. G. (1969). *The Archetypes and the Collective Unconscious* (Collected Works Vol. 9 Part 1). Princeton University Press.
  - [49] Abbott, R. et al. (LIGO Scientific, Virgo, and KAGRA Collaborations) (2021). *GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run*. arXiv:2111.03606 [gr-qc].
  - [50] Abbott, R. et al. (LIGO Scientific, Virgo, and KAGRA Collaborations) (2021). *Tests of General Relativity with GWTC-3* <148>. arXiv:2112.06861 [gr-qc].
  - [51] Lev, F. M. (2010). Introduction to A Quantum Theory over A Galois Field. *Symmetry*, 2(4), 1810–1845.
  - [52] Milne, J. S. (2015). The Riemann Hypothesis over Finite Fields: From Weil to the Present Day (a conjecture resolved for function fields by Deligne). In *The Legacy of Bernhard Riemann After One Hundred and Fifty Years*. International Press.
  - [53] NASA Exoplanet Science Institute (2025). *NASA Exoplanet Archive*. California Institute of Technology.
  - [54] Nielsen, M. A., & Chuang, I. L. (2010). *Quantum Computation and Quantum Information* (10th Anniversary ed.). Cambridge University Press.
  - [55] Oncescu, C.-A., Morwani, D., Jelassi, S., Meterez, A., Kwun, M. & Kakade, S. (2026). The Recurrent Transformer: Greater Effective Depth and Efficient Decoding. arXiv:2604.21215 [cs.LG].

- 
- [56] Particle Data Group et al. (2025). *Review of Particle Physics*. (Includes Monte Carlo Particle Numbering Scheme).
  - [57] Bognár, János (1974). *Indefinite Inner Product Spaces*. Springer Science & Business Media.
  - [58] Susskind, L. (2018). *ER=EPR, GHZ, and the Consistency of Quantum Measurements*. *Phys. Rev. D*.
  - [59] Penrose, R. (1974). The role of aesthetics in pure and applied mathematical research. *Bulletin of the Institute of Mathematics and its Applications*, 10, 266–271.
  - [60] Russell, W. (1926). *The Universal One: Harmonic Nuclear States and Periodic Octaves*.
  - [61] Russell, W. (1926). *The Universal One*. University of Science and Philosophy.
  - [62] Barrett, T.W. (2000). *NASA TelePlasma: Transforming ordinary electromagnetic fields into specially conditioned nonabelian gauge fields*. NASA internal report / BSEI Report 1-97.
  - [63] Fröhlich, H. (1968). “Long-range coherence and energy storage in biological systems.” *Int. J. Quantum Chem.* 2(5), 641–649.
  - [64] Titius, J. D. (1766). *Translation of Charles Bonnet’s Contemplation of Nature*.
  - [65] Nicastro, F. et al. (2018). *Confirming the Detection of two WHIM Systems along the Line of Sight to 1ES 1553+113*. arXiv:1811.03498 [astro-ph.CO].
  - [66] S. Aaronson, “The Busy Beaver Frontier,” *ACM SIGACT News* 51(3), 32–54 (2020).
  - [67] K. J. Arrow, “A Difficulty in the Concept of Social Welfare,” *Journal of Political Economy*, vol. 58, no. 4, pp. 328–346, 1950.
  - [68] S. Atasoy et al., “Connectome-harmonic decomposition of human brain activity reveals dynamical repertoire re-organization under LSD,” *Scientific Reports* 7, 17661 (2017).
  - [69] Atiyah, M. F. & Singer, I. M. (1963). The index of elliptic operators on compact manifolds. *Bull. Amer. Math. Soc.*, 69, 422–433.
  - [70] Zheng, K. et al. (2026). AutoformBot: Multi-agent textbook formalization at scale. arXiv:2605.14332.
  - [71] Baez, J. C. (2002). The Octonions. *Bull. Amer. Math. Soc.*, 39(2), 145–205.
  - [72] S. Bahamonde et al., “Teleparallel Gravity: From Theory to Cosmology,” *Rep. Prog. Phys.* **86**, 026901, 2023.
  - [73] Bender, C. M., Berry, M. V. & Mandilara, A. (2002). Generalized  $\mathcal{PT}$  symmetry and real spectra. *Journal of Physics A: Mathematical and General*, 35(31), L467–L471.
  - [74] C. M. Bender, M. V. Berry, and A. Mandilara, “Generalized  $\mathcal{PT}$  symmetry and real spectra,” *J. Phys. A* 35(31), L467–L471 (2002).

- 
- [75] Bender, C. M., Brody, D. C. & Müller, M. P. (2017). Hamiltonian for the Zeros of the Riemann Zeta Function. *Physical Review Letters*, 118(13), 130201.
  - [76] C. M. Bender, D. C. Brody, and M. P. Müller, “Hamiltonian for the Zeros of the Riemann Zeta Function,” *Physical Review Letters* 118(13), 130201 (2017).
  - [77] F. Black and M. Scholes, “The Pricing of Options and Corporate Liabilities,” *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973.
  - [78] M. A. Blagg, “On a suggested substitute for Bode’s Law,” *MNRAS* 73, 414–422 (1913).
  - [79] Bohm, D. (1980). *Wholeness and the Implicate Order*. Routledge.
  - [80] Borchers, R. E. (1992). Monstrous moonshine and monstrous Lie superalgebras. *Inventiones Mathematicae*, 109(1), 405–444.
  - [81] Bostrom, N. (2011). Infinite Ethics. *Analysis and Metaphysics*, 10, 9–59.
  - [82] T. Bovaird and C. H. Lineweaver, “Exoplanet predictions based on the generalized Titius-Bode relation,” *MNRAS* 435, 1126–1138 (2013).
  - [83] T. H. Boyer, “Quantum Electromagnetic Zero-Point Energy of a Conducting Spherical Shell and the Casimir Model for a Charged Particle,” *Physical Review*, vol. 174, pp. 1764–1776, 1968.
  - [84] Buzzard, K. (2020). Formalising Mathematics in Lean. *Notices of the AMS*, 67(11), 1613–1618.
  - [85] Calaque, D. and Ronchi, S. (2026). Shifted Symplectic Geometry by Examples. arXiv:2510.04625v3.
  - [86] Calude, C. S. & Jürgensen, H. (2005). Is complexity a source of incompleteness? *Advances in Applied Mathematics*, 35(1), 1–15.
  - [87] R. L. Carhart-Harris and K. J. Friston, “REBUS and the Anarchic Brain: Toward a Unified Model of the Brain Action of Psychedelics,” *Pharmacol Rev* 71(3), 316–344 (2019).
  - [88] H. B. G. Casimir, “On the Attraction Between Two Perfectly Conducting Plates,” *Proc. Kon. Ned. Akad. Wetensch.*, vol. 51, pp. 793–795, 1948.
  - [89] Chaitin, G. J. (1975). A theory of program size formally identical to information theory. *J. ACM*, 22(3), 329–340.
  - [90] Chevalley, C. (1936). La théorie du symbole de restes normiques. *Journal für die reine und angewandte Mathematik*, 1936(176), 73–94.
  - [91] F. Brandt, V. Conitzer, U. Endriss, J. Lang, and A. D. Procaccia, eds., *Handbook of Computational Social Choice*, Cambridge University Press, 2016.
  - [92] Connes, A. (1994). *Noncommutative Geometry*. Academic Press.



- 
- [93] Connes, A., Consani, C., & Moscovici, H. (2023). Zeta zeros and prolate wave operators. *arXiv:2310.18223*.
  - [94] A. Connes and C. Consani, “The Arithmetic Site,” *Comptes Rendus Mathématique* 352(12), 971–975 (2014).
  - [95] Connes, A. & Marcolli, M. (2008). *Noncommutative Geometry, Quantum Fields and Motives*. AMS.
  - [96] Connes, A. & Consani, C. (2021). Weil positivity and trace formula. *arXiv:2112.07565*.
  - [97] Conrey, J. B. (2005). Notes on  $L$ -functions and Random Matrix Theory. *Proceedings of Symposia in Pure Mathematics*, 73, 107–130.
  - [98] J. B. Conrey, “Notes on  $L$ -functions and Random Matrix Theory,” *Proc. Sympos. Pure Math.* 73, 107–130 (2005).
  - [99] Conway, J. H. (1976). *On Numbers and Games*. Academic Press.
  - [100] Conway, J. H. & Norton, S. P. (1979). Monstrous Moonshine. *Bull. London Math. Soc.*, 11(3), 308–339.
  - [101] D. Coppersmith, “An approximate Fourier transform,” IBM Research Report RC19642 (1994).
  - [102] M. Creutz, “Monte Carlo study of quantized  $SU(2)$  gauge theory,” *Phys. Rev. D* 21, 2308–2315 (1980).
  - [103] Howard, W. A. (1980). The formulae-as-types notion of construction. In *To H.B. Curry: Essays on Combinatory Logic*, pp. 479–490.
  - [104] T. Damour, “The Problem of Motion in Newtonian and Einsteinian Gravity,” in *Three Hundred Years of Gravitation*, eds. S. Hawking and W. Israel, Cambridge Univ. Press, 1987.
  - [105] de Shalit, E. (2016). Mahler bases and elementary  $p$ -adic analysis. *Journal de Théorie des Nombres de Bordeaux*, 28(3), 597–620.
  - [106] de Moura, L. & Ullrich, S. (2021). The Lean Theorem Prover and Programming Language. In *CADE-28*, LNCS 12699, pp. 625–635.
  - [107] de Rham, G. (1931). Sur l’analyse situs des variétés à  $n$  dimensions. *J. Math. Pures Appl.*, 10, 115–200.
  - [108] S. F. Dermott, “On the origin of commensurabilities in the solar system—II,” *MNRAS* 141, 363–376 (1968).
  - [109] de Shalit, E. (2016). Mahler bases and elementary  $p$ -adic analysis. *Journal de Théorie des Nombres de Bordeaux*, 28(3), 597–620.
  - [110] E. de Shalit, “Mahler bases and elementary  $p$ -adic analysis,” *J. Théorie des Nombres de Bordeaux* 28(3), 597–620 (2016).



- 
- [111] H. T. Diep, ed., *Frustrated Spin Systems*, 2nd ed., World Scientific, 2013.
  - [112] Dirac, P. A. M. (1930). *The Principles of Quantum Mechanics*. Oxford University Press.
  - [113] F. Eckerli and J. Osterrieder, “Generative Adversarial Networks in Finance: An Overview,” *arXiv:2106.06364*, 2021.
  - [114] J. Eichenauer and J. Lehn, *A Non-Linear Congruential Pseudo Random Number Generator*, *Statistische Hefte*, 27(1):315–326, 1986.
  - [115] E. Eichten, K. Gottfried, T. Kinoshita, K. Lane, and T.-M. Yan, “Charmonium: Comparison with Experiment,” *Phys. Rev. D* **21**, 203, 1980.
  - [116] V. Elser and C. L. Henley, “Crystal and quasicrystal structures in Al–Mn alloys,” *Phys. Rev. Lett.* **55**, 2883–2886 (1985).
  - [117] Euler, L. (1748). *Introductio in Analysin Infinitorum*. Lausanne.
  - [118] J. D. Fernstrom and R. J. Wurtman, “Brain serotonin content: physiological regulation by plasma neutral amino acids,” *Science* **173**, 149 (1971).
  - [119] Feynman, R. P., Leighton, R. B., & Sands, M. (1964). *The Feynman Lectures on Physics*. Addison-Wesley.
  - [120] R. P. Feynman, “The behavior of hadron collisions at extreme energies,” in *High Energy Collisions: Third International Conference*, Gordon and Breach, 1969.
  - [121] Fibonacci (Leonardo of Pisa). (1202). *Liber Abaci*. Translated by L.E. Sigler (2002), Springer.
  - [122] Man AHL Research, “Fin-GAN: Forecasting and Simulating Financial Time Series with Generative Adversarial Networks,” *Man Group Research Papers*, 2024.
  - [123] S. Buzsáki and K. Mizuseki, “The log-dynamic brain: how skewed distributions affect network operations,” *Nature Reviews Neuroscience* **15**, 264–278 (2014).
  - [124] E. Bullmore et al., “Fractal analysis of the electroencephalogram in depression and schizophrenia,” *IEEE Trans Biomed Eng* (1994).
  - [125] J. Friedel, “Do Hume-Rothery rules still hold? Old and new electron phases in metallic alloys,” *Phil. Mag. B* **58**(4), 413–427 (1988).
  - [126] D. Gabor, “Theory of Communication,” *Journal of the Institution of Electrical Engineers*, vol. 93, no. 26, pp. 429–457, 1946.
  - [127] M. Gell-Mann, “A schematic model of baryons and mesons,” *Physics Letters*, vol. 8, no. 3, pp. 214–215, 1964.
  - [128] A. Gibbard, “Manipulation of Voting Schemes: A General Result,” *Econometrica*, vol. 41, no. 4, pp. 587–601, 1973.

- 
- [129] Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38, 173–198.
  - [130] Goodstein, R. L. (1944). On the restricted ordinal theorem. *J. Symbolic Logic*, 9(2), 33–41.
  - [131] R. L. Goodstein, “Transfinite Ordinals in Recursive Number Theory,” *Journal of Symbolic Logic* 12(4), 123–129 (1947).
  - [132] F. Graner and B. Dubrulle, “Titius-Bode laws in the solar system. I. Scale invariance explains everything,” *Astron. Astrophys.* 282, 262–268 (1994).
  - [133] J. Greensite, *An Introduction to the Confinement Problem*, 2nd ed., Lecture Notes in Physics, vol. 972, Springer, 2020. (Center symmetry and  $Z(N)$  vortex confinement: Ch. 4–5.)
  - [134] Grothendieck, A. (1984). *Esquisse d’un Programme*.
  - [135] Grothendieck, A. (1985–1986). *Récoltes et Semailles: Réflexions et Témoignage sur un Passé de Mathématicien*. Université des Sciences et Techniques du Languedoc, Montpellier.
  - [136] A. Grothendieck and J. Dieudonné, “Éléments de géométrie algébrique,” *Inst. Hautes Études Sci. Publ. Math.*, 1960–1967.
  - [137] Gutin, R. (2020). Constructive proof of Herschfeld’s Convergence Theorem. *Mathematics of Computation*, 89, 237–251.
  - [138] S. Hameroff and R. Penrose, “Consciousness in the universe: A review of the ‘Orch OR’ theory,” *Physics of Life Reviews* 11(1), 39–78 (2014).
  - [139] Hatcher, A. (2002). *Algebraic Topology*. Cambridge University Press.
  - [140] S. Hod (1998). “Bohr’s correspondence principle and the area spectrum of quantum black holes.” *Phys. Rev. Lett.* 81, 4293.
  - [141] Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proc. Natl. Acad. Sci.*, 79(8), 2554–2558.
  - [142] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th ed., Pearson, 2018.
  - [143] K. Ilinski, “Physics of Finance,” *arXiv:hep-th/9710148*, 1997.
  - [144] A. Jaffe and E. Witten, “Quantum Yang-Mills Theory,” Official Problem Description, Clay Mathematics Institute (2000).
  - [145] Jaramillo Salazar, J. D. (2020). Hyperoperations in Exponential Fields. *Journal of Algebra and Its Applications*, 19(1).
  - [146] Q.-D. Jiang and F. Wilczek, “Chiral Casimir forces: Repulsive, enhanced, tunable,” *Physical Review B*, vol. 99, no. 12, p. 125403, 2019.

- 
- [147] M. Aker *et al.* (KATRIN Collaboration), “Direct neutrino-mass measurement with sub-electronvolt sensitivity,” *Nature Physics*, vol. 18, pp. 160–166, 2022.
  - [148] Lurie, J. (2018–). *Kerodon*. <https://kerodon.net>.
  - [149] J. Kim, “Neutrino Mass and Mixing,” *Phys. Rep.* **150**, 1–177, 2001.
  - [150] Knuth, D. E. (1974). *Surreal Numbers*. Addison-Wesley.
  - [151] D. E. Knuth, “Mathematics and Computer Science: Coping with Finiteness,” *Science* 194(4271), 1235–1242 (1976). Introduces the up-arrow notation for Goodstein’s hyperoperation hierarchy.
  - [152] A. N. Kolmogorov, “The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers,” *Doklady Akademii Nauk SSSR*, vol. 30, pp. 299–303, 1941.
  - [153] K. R. Konkoly et al., “Real-time dialogue between experimenters and dreamers during REM sleep,” *Current Biology* 31(7), 1417–1427.e6 (2021).
  - [154] M. Farach-Colton, A. Krapivin, and W. Kuszmaul, “Optimal Bounds for Open Addressing Without Reordering,” arXiv preprint arXiv:2501.02305 (2025).
  - [155] Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3), 183–191.
  - [156] D. Layden et al., “Quantum Error Correction Below the Surface Code Threshold,” *Nature* **638**, 2025.
  - [157] J.-L. Le Mouél, V. Courtillot, and S. Gibert, “SolDynamo: Solar Dynamo and Planetary Coupling,” preprint, 2025.
  - [158] Lev, F. M. (2020). Finite Mathematics as the Foundation of Classical Mathematics and Quantum Theory. Springer.
  - [159] Lev, F. M. (2010). Introduction to A Quantum Theory over A Galois Field. *Symmetry*, 2(4), 1810–1845.
  - [160] F. M. Lev, “Introduction to A Quantum Theory over A Galois Field,” *Symmetry* 2(4), 1810–1845 (2010).
  - [161] Lockwood, J. (2024). Chronometric Coordination in Distributed Temporal Systems. *Journal of Distributed Computing*, 37(4), 215–234.
  - [162] J. Lockwood, “Fractal Spacetime and Zero-Point Energy,” 2025.
  - [163] M. López de Prado, *Advances in Financial Machine Learning*, Wiley, 2018. Chapter 11: “The Dangers of Backtesting.” Sustained precision above 50% on crash classification tasks invites scrutiny for look-ahead bias and data leakage.

- 
- [164] J.-P. Luminet, J. R. Weeks, A. Riazuelo, R. Lehoucq, J.-P. Uzan (2003). “Dodecahedral space topology as an explanation for weak wide-angle temperature correlations in the cosmic microwave background.” *Nature* 425, 593–595.
  - [165] Lurie, J. (2009). *Higher Topos Theory*. Princeton University Press.
  - [166] Mac Lane, S. & Moerdijk, I. (1992). *Sheaves in Geometry and Logic*. Springer-Verlag.
  - [167] Mal’cev, A. I. (1955). Analytic loops. *Mat. Sb.*, 36(78), 569–576.
  - [168] J. Maluf, “The Teleparallel Equivalent of General Relativity,” *Annalen der Physik* **525**, 339–357, 2013.
  - [169] Mandelbrot, B. B. (1982). *The Fractal Geometry of Nature*. W.H. Freeman.
  - [170] B. B. Mandelbrot, “The Variation of Certain Speculative Prices,” *The Journal of Business*, vol. 36, no. 4, pp. 394–419, 1963.
  - [171] N. S. Manton, “Fluid mechanical models of financial markets,” Working paper, DAMTP Cambridge, 2002.
  - [172] Massot, P. (2023). Lean Blueprints: Bridging Informal and Formal Mathematics. In *ITP 2023*, LIPIcs.
  - [173] The mathlib Community. (2020). The Lean Mathematical Library. In *CPP ’20*, pp. 367–381. ACM.
  - [174] May, J. P. & Ponto, K. (2012). *More Concise Algebraic Topology*. University of Chicago Press.
  - [175] J. McKinney et al., “A revised mechanism for tryptophan hydroxylase,” *J. Biol. Chem.* 280, 13265 (2005).
  - [176] Milne, J. S. (2015). The Riemann Hypothesis over Finite Fields: From Weil to the Present Day (a conjecture resolved for function fields by Deligne). In *The Legacy of Bernhard Riemann After One Hundred and Fifty Years*. International Press.
  - [177] J. S. Milne, “The Riemann Hypothesis over Finite Fields,” in *The Legacy of Bernhard Riemann After One Hundred and Fifty Years*, International Press (2015).
  - [178] J. S. Milne, “The Riemann Hypothesis over Finite Fields: From Weil to the Present Day,” in *The Legacy of Bernhard Riemann After One Hundred and Fifty Years*, International Press (2015).
  - [179] J. S. Milne, “Class Field Theory,” lecture notes, v4.03, 2020.
  - [180] Milnor, J. (1963). *Morse Theory*. Princeton University Press.
  - [181] Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. *Analytic Number Theory*, 181–193.

- 
- [182] L. Motl and A. Neitzke (2003). “Asymptotic black hole quasinormal frequencies.” *Adv. Theor. Math. Phys.* 7, 307–330.
  - [183] R. A. E. Edden et al., “Proton MR spectroscopy of GABA, glutamate, and glutamine,” *Neuroimage*, 2012.
  - [184] A. Bogaschewsky et al., “Reproducibility of Glx and GABA measurements in the human brain,” *Magnetic Resonance in Medicine*, 2019.
  - [185] T. Nagatsu, M. Levitt, and S. Udenfriend, “Tyrosine Hydroxylase: The Initial Step in Norepinephrine Biosynthesis,” *J Biol Chem* 239, 2910–2917 (1964).
  - [186] Nash, J. F. (1950). Equilibrium points in  $n$ -person games. *Proc. Natl. Acad. Sci.*, 36(1), 48–49.
  - [187] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften, vol. 322, Springer, 1999.
  - [188] I. Newton, *Opticks: or, A Treatise of the Reflexions, Refractions, Inflexions and Colours of Light*, 1704.
  - [189] Odlyzko, A. M. (1987). On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation*, 48(177), 273–308.
  - [190] Odrzywolek, A. (2026). All elementary functions from a single operator. *arXiv preprint*.
  - [191] A. Khrennikov, *Information Dynamics in Cognitive, Psychological, Social and Anomalous Phenomena*, Springer (2004).
  - [192] Penrose, R. (1989). *The Emperor’s New Mind*. Oxford University Press.
  - [193] Penrose, R. (1974). The role of aesthetics in pure and applied mathematical research. *Bull. IMA*, 10(7/8), 266–271.
  - [194] Ken Perlin, *An Image Synthesizer*, SIGGRAPH Computer Graphics, 19(3):287–296, 1985.
  - [195] M. Persson et al., *Minecraft: Deterministic Voxel Generation and LCGs*, Mojang Specifications (Informal Technical Documentation), 2009–2011.
  - [196] M. Pitkänen, “TGD Inspired Theory of Consciousness,” *Journal of Non-Locality and Remote Mental Interactions* 1 (2002).
  - [197] Planck Collaboration (2020). “Planck 2018 results. VII. Isotropy and statistics of the CMB.” *Astron. Astrophys.* 641, A7.
  - [198] J. M. Pollard, “The fast Fourier transform in a finite field,” *Math. Comp.* 25(114), 365–374 (1971).
  - [199] Pantev, T., Toën, B., Vaquié, M. and Vezzosi, G. (2013). Shifted symplectic structures. *Publ. Math. Inst. Hautes Études Sci.* 117, 271–328.

- 
- [200] T. Radó, “On Non-Computable Functions,” *Bell System Technical Journal* 41(3), 877–884 (1962).
  - [201] Riemann, B. (1859). Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte der Berliner Akademie*.
  - [202] Robinson, A. (1966). *Non-standard Analysis*. North-Holland.
  - [203] A. E. Roy and M. W. Ovenden, “On the occurrence of commensurable mean motions in the solar system,” *MNRAS* 114, 232–241 (1954).
  - [204] Xin, H. et al. (2025). Safe: Step-aware formal verification for LLM reasoning. *Proc. ACL*, pp. 4821–4838.
  - [205] M. A. Satterthwaite, “Strategy-Proofness and Arrow’s Conditions,” *Journal of Economic Theory*, vol. 10, no. 2, pp. 187–217, 1975.
  - [206] J. Savitz, “The kynurenine pathway: a finger in every pie,” *Mol Psychiatry* 25, 131–147 (2020).
  - [207] Schafer, R. D. (1966). *An Introduction to Nonassociative Algebras*. Academic Press.
  - [208] E. Schrödinger, “Quantisierung als Eigenwertproblem,” *Annalen der Physik*, vol. 384, no. 4, pp. 361–376, 1926.
  - [209] E. Schrödinger, *What is Life? The Physical Aspect of the Living Cell*, Cambridge University Press, 1944. Chapter 6: “Order, Disorder, and Entropy.”
  - [210] L. Schwieler et al., “Electroconvulsive therapy suppresses the neurotoxic branch of the kynurenine pathway in treatment-resistant depressed patients,” *Transl Psychiatry* 6, e906 (2016).
  - [211] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics, vol. 67, Springer, 1979.
  - [212] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
  - [213] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal* 27, 379–423, 623–656 (1948).
  - [214] P. W. Shor, “Algorithms for quantum computation: discrete logarithms and factoring,” *Proceedings 35th Annual Symposium on Foundations of Computer Science*, IEEE, 124–134 (1994).
  - [215] A. T. Shulgin and A. Shulgin, *PiHKAL: A Chemical Love Story*, Transform Press, 1991.
  - [216] M. Sono et al., “Heme-containing oxygenases,” *Chem. Rev.* 96, 2841 (1996).
  - [217] B. Srinivasan et al., “TEER measurement techniques for in vitro barrier model systems,” *JALA* 20, 107 (2015).



- 
- [218] Steenrod, N. (1951). *The Topology of Fibre Bundles*. Princeton University Press.
  - [219] F. Stillinger, “Axiomatic Basis for Spaces with Noninteger Dimension,” *J. Math. Phys.* **18**, 1224–1234, 1977.
  - [220] J. Stöckigt et al., “The Pictet–Spengler reaction in nature and in organic chemistry,” *Angew. Chem.* **50**, 7580 (2011).
  - [221] S. Takahashi, Y. Chen, and K. Tanaka-Ishii, “Modeling Financial Time-Series with Generative Adversarial Networks,” *Physica A: Statistical Mechanics and its Applications*, vol. 527, 121261, 2019.
  - [222] Tarski, A. (1936). Der Wahrheitsbegriff in den formalisierten Sprachen. *Studia Philosophica*, **1**, 261–405.
  - [223] G. ’t Hooft, “Topology of the gauge condition and new confinement phases in non-Abelian gauge theories,” *Nucl. Phys. B* **190**, 455–478 (1981).
  - [224] J. Yoon, D. Jarrett, and M. van der Schaar, “Time-series Generative Adversarial Networks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
  - [225] Eby, J., Safronova, M. S., & Tsai, Y.-D. (2022). Direct detection of ultralight dark matter bound to the Sun with space quantum sensors. *Nature Astronomy*, **6**, 1233–1239.
  - [226] X. Li, V. Metsis, H. Wang, and A. Ngu, “TTS-GAN: A Transformer-based Time-Series Generative Adversarial Network,” *arXiv:2202.02691v3*, 2024.
  - [227] G. Varieschi, “Newtonian Fractional-Dimension Gravity and MOND,” *Found. Phys.* **50**, 1608–1644, 2020.
  - [228] Veblen, O. (1908). Continuous Increasing Functions of Finite and Transfinite Ordinals. *Trans. AMS*, **9**(3), 280–292.
  - [229] Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press.
  - [230] F. Wilczek, “Ultraviolet Stability of the Vacuum in a Non-Abelian Gauge Theory,” *Phys. Rev. Lett.* **30**, 1343–1346, 1973.
  - [231] K. G. Wilson, “Confinement of quarks,” *Phys. Rev. D* **10**, 2445–2459 (1974).
  - [232] La Rue, D. (2026). Aingel\_proofs: The ZKP Holochain Envelope over the Metareal Manifold. Formal algebraic synthesis.
  - [233] La Rue, D. (2026). *The Golden Tetrahedron: Gauge Center Hierarchy and Chronometric Structure*.
  - [234] La Rue, D. (2026). InfochemicalTime\_proofs: Quasi-Epi-Periodic Geometric Time. Formal algebraic synthesis.



- 
- [235] La Rue, D. (2026). *MetarealManifold\_proofs: The 12-Dimensional Observer-Manifold Space*. Formal algebraic synthesis.
  - [236] La Rue, D. (2026). *PrimeFieldMechanics\_proofs: Agentic Mechanics*. Formal algebraic synthesis.
  - [237] D. La Rue, “The Bionetic Ambrosia Protocol,” June 2026.
  - [238] Various, “Bitcoin as a Leading Indicator for Risk Appetite: 24/7 Liquidity Signals and Equity Market Correlation,” Market microstructure research and institutional reports, 2024–2026.
  - [239] Various, “Stock Market Crash Prediction: Evaluation Metrics and Model Comparison,” Academic surveys across machine learning and econometric approaches, 2020–2024. Typical precision for 3-month crash prediction:  $\sim 15\%$  at recall  $\sim 59\%$ .
  - [240] D. La Rue, “Digital Wave Mechanics,” 2025.
  - [241] J. Lockwood and D. La Rue, “Digital Wave Mechanics and Fractal Spacetime: Adelic Orbit Structure of the Chronometric Triangle,” preprint, June 2026.
  - [242] Various, “Machine Learning Early Warning Systems for Financial Crises,” Studies from Bank of Canada, ECB, IMF, and academic papers, 2020–2026. State-of-the-art AUROC: 0.75–0.88 for 6-month recession forecasting using 15–50 macroeconomic indicators.
  - [243] Various, “GAN-Guided Diffusion Models for Financial Time Series,” *arXiv preprints*, 2024–2025.
  - [244] D. La Rue, “The Golden Tetrahedron: Algebraic Foundations of Standard Model Symmetries,” 2026.
  - [245] Various, “Human-Computer Interaction in Financial Trading Systems: Market Research and Design Principles,” Industry reports and academic surveys, 2025–2026.
  - [246] La Rue, D. (2026). *Protoreal Zeta*. Formal verification of topological and algebraic axioms. <https://github.com/Dielawn-01/ProtorealZeta>.
  - [247] La Rue, D. & Lockwood, J. (2026). *SU(3) Center Drift Extension: Golden Cosets and Temporal Spectral Dynamics*.
  - [248] D. La Rue. *The SU(3) Center Triangle and Fractal SpaceTime* (Digital Wave Mechanics). Spectrality Institute, 2026.
  - [249] D. La Rue. *Logical Creativity and the La Rue Manifold*. Spectrality Institute, 2026.
  - [250] D. La Rue, “The SU(3) Center Triangle: Golden Split Primes, Standing Waves in Digit Space, and the Dark-Bright Palindrome Resonance,” preprint, June 2026.
  - [251] D. La Rue and J. Lockwood, “Digital Wave Mechanics and Fractal Spacetime,” Preprint, June 2026.

- 
- [252] D. La Rue, “Logical Creativity: A Discussion on the Foundations of Mathematics,” preprint, June 2026.
  - [253] D. La Rue, *Logical Creativity*, Protoreal Zeta Series, 2026.
  - [254] D. La Rue and J. Lockwood. *Digital Wave Mechanics*. Spectrality Institute, 2025.
  - [255] J. Lockwood. *Zero-Point Energy: First Principles, Units, and Observables*. Spectrality Institute, Sept 2025.
  - [256] J. Lockwood, “Chronometric Constants and Frozen Spectral Relations,” Private communication, 2026.
  - [257] J. Lockwood, “Chronometric Constants and Frozen Spectral Relations (Chrono 3),” private communication, 2026.
  - [258] J. Lockwood, “Chronometric Constants and Frozen Spectral Relations (Chrono 4),” private communication, 2026.
  - [259] J. Lockwood and D. La Rue, “Digital Wave Mechanics: From Newton’s Prism to Standing Waves in Finite Golden Fields,” 2026.
  - [260] M. Lockwood, “Solar Cycle Prediction: v11 Update,” private communication, 2025.
  - [261] D. La Rue, “Metareal ASI Architecture: The Golden Veblen Resolution of Gödelian Incompleteness,” June 2026.
  - [262] D. La Rue, *InfoPhys Lattice Dynamics*, Metamaterial Research Export, 2026. <https://github.com/Dielawn-01/InfoPhys>
  - [263] D. La Rue, *Protoreal Zeta*, formal axioms, 2026. [https://github.com/Dielawn-01/Protoreal\\_Zeta](https://github.com/Dielawn-01/Protoreal_Zeta)
  - [264] J. Lockwood and D. Hansley, *Royal Jelly: Field–Transport Foundations for Bioelectromagnetics*, Spectrality Institute, 2025.
  - [265] D. La Rue, *Transport–Algebraic Correspondence in Bioelectromagnetics: An Analysis of the Royal Jelly Framework*, InfoPhys Axioms, 2026.
  - [266] Steiniger, R., “Engineering Persistent Geometric Identities in LLMs,” Preprint v3, May 2026.
  - [267] M. Walker, “The Infoton, the Blackbody Constant of Information-Energy, and the Landauer Wall,” private communication, 2024.
  - [268] F. Wilczek, “Quantum Time Crystals,” *Physical Review Letters*, vol. 109, 160401, 2012.
  - [269] H. Watanabe and M. Oshikawa, “Absence of Quantum Time Crystals,” *Physical Review Letters*, vol. 114, 251603, 2015.

- 
- [270] J. Zhang et al., “Observation of a Discrete Time Crystal,” *Nature*, vol. 543, pp. 217–220, 2017.
  - [271] X. Mi et al., “Time-crystalline eigenstate order on a quantum processor,” *Nature*, vol. 601, pp. 531–536, 2022.
  - [272] D. V. Else, B. Bauer, and C. Nayak, “Floquet Time Crystals,” *Physical Review Letters*, vol. 117, 090402, 2016.
  - [273] C. R. Cowley, W. P. Bidelman, S. Hubrig, G. Mathys, and D. J. Bord, “On the Possible Presence of Promethium in the Spectrum of HD 101065 (Przybylski’s Star),” *The Astrophysical Journal*, vol. 603, pp. 354–367, 2004.
  - [274] V. F. Gopka, A. V. Yushchenko, A. V. Shavrina et al., “Identification of absorption lines of the transuranium elements in the spectrum of Przybylski’s star (HD 101065),” *Kinematika i Fizika Nebesnykh Tel*, vol. 20, pp. 210–220, 2004.
  - [275] F. Delbaen and W. Schachermayer, “A general version of the fundamental theorem of asset pricing,” *Mathematische Annalen*, vol. 300, pp. 463–520, 1994.
  - [276] R. D. Wyckoff, *The Richard D. Wyckoff Method of Trading and Investing in Stocks*. New York: Wyckoff Associates, 1931.
  - [277] A. Weil, *Sur les courbes algébriques et les variétés qui s’en déduisent*. Paris: Hermann, 1948.
  - [278] J. M. Harrison and D. M. Kreps, “Martingales and arbitrage in multiperiod securities markets,” *Journal of Economic Theory*, vol. 20, no. 3, pp. 381–408, 1979.
  - [279] J. M. Harrison and S. R. Pliska, “Martingales and stochastic integrals in the theory of continuous trading,” *Stochastic Processes and their Applications*, vol. 11, no. 3, pp. 215–260, 1981.
  - [280] J. L. Doob, *Stochastic Processes*, Wiley, 1953. Chapter VII: Martingale theory and the tower property of conditional expectations.
  - [281] H. Niederreiter, *Random Number Generation and Quasi-Monte Carlo Methods*, CBMS-NSF Regional Conference Series in Applied Mathematics, vol. 63, SIAM, 1992. Chapters 8–9: inversive congruential generators and their equidistribution properties.
  - [282] M. Avellaneda and J.-H. Lee, “Statistical Arbitrage in the U.S. Equities Market,” *Quantitative Finance*, vol. 10, no. 7, pp. 761–782, 2010.
  - [283] B. Hurst, Y. H. Ooi, and L. H. Pedersen, “A Century of Evidence on Trend-Following Investing,” *Journal of Portfolio Management*, vol. 44, no. 1, pp. 15–29, 2017.
  - [284] E. E. Qian, “Risk Parity Portfolios: Efficient Portfolios Through True Diversification,” PanAgora Asset Management, White Paper, 2005.

- [285] P. Embrechts, A. McNeil, and D. Straumann, “Correlation and Dependence in Risk Management: Properties and Pitfalls,” in *Risk Management: Value at Risk and Beyond*, ed. M. Dempster, Cambridge University Press, pp. 176–223, 2002.
- [286] R. Penrose, “The Role of Aesthetics in Pure and Applied Mathematical Research,” *Bulletin of the Institute of Mathematics and its Applications*, vol. 10, no. 7/8, pp. 266–271, 1974. *Introduces aperiodic tilings with golden-ratio structure (Penrose tilings), the spatial precursor to quasicrystals.*
- [287] E. Schrödinger, *What is Life? The Physical Aspect of the Living Cell*, Cambridge University Press, 1944. *Predicts that the genetic material is an “aperiodic crystal”—a structure with long-range order but no periodic repetition.*
- [288] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, “Metallic Phase with Long-Range Orientational Order and No Translational Symmetry,” *Physical Review Letters*, vol. 53, no. 20, pp. 1951–1953, 1984. *Discovery of quasicrystals. Nobel Prize in Chemistry, 2011.*
- [289] F. H. C. Crick, “Codon–Anticodon Pairing: The Wobble Hypothesis,” *Journal of Molecular Biology*, vol. 19, no. 2, pp. 548–555, 1966. *The wobble position allows non-canonical base pairing at the third codon position, introducing controlled imperfection into the genetic code.*
- [290] N. Y. Yao, A. C. Potter, I.-D. Potirniche, and A. Vishwanath, “Discrete Time Crystals: Rigidity, Criticality, and Realizations,” *Physical Review Letters*, vol. 118, no. 3, p. 030401, 2017. *Theoretical blueprint for discrete time crystals, defining conditions for robust subharmonic order in Floquet systems.*
- [291] J.-L. Le Mouél, F. Lopes, V. Courtillot, and D. Gibert, “Planetary Forcing of Solar Activity: A Review,” preprint (SolDynamoExtForcing), 2025. *Identifies 11 shared pseudo-periods between sunspot number (SSN, 1750–2025) and length-of-day (LOD, 1780–2025) via Singular Spectrum Analysis. Correlation  $r > 0.999$ , bootstrap  $p \leq 10^{-3}$ .*
- [292] R. Holme and O. de Viron, “Characterization and implications of intradecadal variations in length of day,” *Nature*, vol. 499, pp. 202–204, 2013. *LOD analysis establishing electromagnetic core-mantle coupling coefficient  $\approx 0.80$  for Earth.*
- [293] A. M. Dziewonski and D. L. Anderson, “Preliminary reference Earth model,” *Physics of the Earth and Planetary Interiors*, vol. 25, no. 4, pp. 297–356, 1981. *PREM. Core radius  $R_{\text{core}} = 3480$  km, planet radius  $R_{\oplus} = 6371$  km, ratio = 0.546.*
- [294] G. H. Pettengill and R. B. Dyce, “A radar determination of the rotation of the planet Mercury,” *Nature*, vol. 206, pp. 1240, 1965. *Discovery of Mercury’s 3:2 spin-orbit resonance. Tidal lock zeroes the conjugate golden channel  $\bar{\varphi} = (\omega - \iota)/2 = 0$ .*
- [295] P. Alken et al., “International Geomagnetic Reference Field: the thirteenth generation,” *Earth, Planets and Space*, vol. 73, art. 49, 2021. *IGRF-13. Earth dipole tilt =  $11.5^\circ \pm 0.3^\circ$ .*
- [296] J. E. P. Connerney et al., “A new model of Jupiter’s magnetic field at the completion of Juno’s prime mission,” *Journal of Geophysical Research: Planets*, vol. 127, e2021JE007055, 2022. *JRM33 model. Jupiter dipole tilt =  $10.3^\circ \pm 0.5^\circ$ .*

- 
- [297] M. K. Dougherty et al., “Saturn’s magnetic field revealed by the Cassini Grand Finale,” *Science*, vol. 362, eaat5434, 2018. *Cassini MAG. Saturn dipole tilt  $< 0.01^\circ$  (axisymmetric), consistent with hexagonal ( $k = 6$ ) symmetry lock.*
  - [298] N. F. Ness, M. H. Acuña, K. W. Behannon, L. F. Burlaga, J. E. P. Connerney, and R. P. Lepping, “Magnetic fields at Uranus,” *Science*, vol. 233, pp. 85–89, 1986. *Voyager 2. Uranus dipole tilt  $58.6^\circ \pm 2^\circ$  from rotation axis, offset 0.3 planetary radii from center.*
  - [299] N. F. Ness, M. H. Acuña, L. F. Burlaga, J. E. P. Connerney, and R. P. Lepping, “Magnetic fields at Neptune,” *Science*, vol. 246, pp. 1473–1478, 1989. *Voyager 2. Neptune dipole tilt  $47^\circ \pm 3^\circ$  from rotation axis, offset 0.55 planetary radii from center.*
  - [300] M. G. Kivelson, K. K. Khurana, and M. Volwerk, “The permanent and inductive magnetic moments of Ganymede,” *Icarus*, vol. 157, pp. 507–522, 2002. *Galileo. Ganymede intrinsic dipole anti-aligned with Jupiter’s field ( $\sim 176^\circ$ ). Only moon with self-sustaining magnetosphere.*
  - [301] H. A. Kramers, “Brownian motion in a field of force and the diffusion model of chemical reactions,” *Physica*, vol. 7, no. 4, pp. 284–304, 1940. *The foundational derivation of the overdamped Kramers escape rate for barrier crossing in the presence of thermal noise.*
  - [302] S. H. Snyder and C. R. Merrill, “A relationship between the hallucinogenic activity of drugs and their electronic configuration,” *Proc. Natl. Acad. Sci. USA*, vol. 54, no. 1, pp. 258–266, 1965. *Pioneering QSAR analysis linking psychedelic potency to HOMO energy and superdelocalizability of the indole  $\pi$ -system.*
  - [303] U.S. Food and Drug Administration, “FDA approves first deuterated drug—Austedo (deutetrabenazine) for Huntington’s disease chorea,” FDA Press Release, April 2017. *First approved deuterium-modified drug. Six  $^1\text{H} \rightarrow ^2\text{D}$  substitutions reduce CYP2D6 metabolism via kinetic isotope effect ( $\text{KIE} \approx 2\text{--}5$ ).*
  - [304] A. A. Albert, “Power-Associative Rings,” *Trans. Amer. Math. Soc.*, vol. 64, no. 3, pp. 552–593, 1948. *Foundational classification of power-associative algebras. Establishes that Jordan algebras and alternative algebras are power-associative.*
  - [305] I. Assem, D. Simson, and A. Skowroński, *Elements of the Representation Theory of Associative Algebras*, London Mathematical Society Student Texts, vol. 65, Cambridge University Press, 2006. *Standard reference for quiver representations and Auslander–Reiten theory.*
  - [306] T. Lin and C. T. Radó, “On Non-Computable Functions and Computability Theory,” *Notices of the AMS*, 1965.
  - [307] M. E. J. Newman, “Modularity and community structure in networks,” *Proc. Natl. Acad. Sci. USA*, vol. 103, no. 23, pp. 8577–8582, 2006.
  - [308] C. T. C. Wall, “Determination of the Cobordism Ring,” *Annals of Mathematics*, vol. 72, no. 2, pp. 292–311, 1960. *Classification of oriented cobordism ring. Foundational for topological field theories.*



- 
- [309] D. J. Watts and S. H. Strogatz, “Collective dynamics of ‘small-world’ networks,” *Nature*, vol. 393, pp. 440–442, 1998.
  - [310] R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183–191, 1961. *Establishes the Landauer limit:  $k_B T \ln 2$  per bit erasure.*
  - [311] J. M. Beggs and D. Plenz, “Neuronal Avalanches in Neocortical Circuits,” *Journal of Neuroscience*, vol. 23, no. 35, pp. 11167–11177, 2003. *Discovery of neuronal avalanches following power-law distributions, evidence for self-organized criticality in cortex.*
  - [312] H. A. Bethe, “Energy Production in Stars,” *Physical Review*, vol. 55, no. 5, pp. 434–456, 1939. *CNO cycle and pp-chain. Nobel Prize in Physics 1967.*
  - [313] D. G. Blackmond, “The Origin of Biological Homochirality,” *Cold Spring Harbor Perspectives in Biology*, vol. 2, a002147, 2010. *Review of chiral symmetry breaking mechanisms in prebiotic chemistry.*
  - [314] D. L. D. Caspar and A. Klug, “Physical Principles in the Construction of Regular Viruses,” *Cold Spring Harbor Symposia on Quantitative Biology*, vol. 27, pp. 1–24, 1962. *Caspar–Klug theory of icosahedral virus capsid structure. Triangulation numbers.*
  - [315] L. Cronin and S. I. Walker, “Beyond prebiotic chemistry,” *Science*, vol. 352, no. 6290, pp. 1174–1175, 2016. *Assembly theory and the quantification of molecular complexity.*
  - [316] S. A. Kauffman, *At Home in the Universe: The Search for Laws of Self-Organization and Complexity*, Oxford University Press, 1995.
  - [317] P. J. Lu and P. J. Steinhardt, “Decagonal and Quasi-Crystalline Tilings in Medieval Islamic Architecture,” *Science*, vol. 315, no. 5815, pp. 1106–1110, 2007.
  - [318] D. La Rue, “Temporal Quasicrystals and the QT-45 Hypothesis,” preprint, 2026.
  - [319] F. Tassinari et al., “Enhanced Hydrogen Production from a Chiral Conducting Polymer,” *ACS Nano*, vol. 13, no. 8, pp. 8845–8852, 2019. *Experimental demonstration of chirality-induced spin selectivity (CISS) effect.*
  - [320] S. Weinberg, *The Quantum Theory of Fields, Volume I: Foundations*, Cambridge University Press, 1995. *Derivation of the  $7/8$  spin-statistics factor for fermionic vacuum energy density (§5.3).*
  - [321] G. S. Engel, T. R. Calhoun, E. L. Read, T.-K. Ahn, T. Mančal, Y.-C. Cheng, R. E. Blankenship, and G. R. Fleming, “Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems,” *Nature*, vol. 446, pp. 782–786, 2007.
  - [322] M. Kohmoto, L. P. Kadanoff, and C. Tang, “Localization problem in one dimension: Mapping and escape,” *Phys. Rev. Lett.*, vol. 50, no. 23, pp. 1870–1872, 1983.



- 
- [323] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, “Metallic Phase with Long-Range Orientational Order and No Translational Symmetry,” *Physical Review Letters*, vol. 53, no. 20, pp. 1951–1953, 1984. *Discovery of quasicrystals. Nobel Prize in Chemistry, 2011.*
  - [324] J. Lockwood, “GMF-34: 34-Gradient Plasma Materials-Synthesis Forge,” schematics v1/v2/v8, private communication, 2026. *34-gradient radial layout with central synthesis volume, MHD governing equations, and 4-band acoustic transducer architecture.*
  - [325] D. La Rue, RussellDiagram.lean: Russell Octave Harmonic Structure in Lean 4, InfoPhysAxioms, 2026. *Formalizes the Russell 12-tone cycle, carbon–silicon umbral shift, and the RussellOctaveNode  $\rightarrow$  ProtorealManifold projection.*
  - [326] U. Mizutani and H. Sato, “The Physics of the Hume-Rothery Electron Concentration Rule,” *Crystals*, vol. 7, no. 1, art. 9, 2017. *Comprehensive derivation of  $e/a = 7/4$  for icosahedral QCs via Friedel’s scattering framework. Establishes the Jones zone / Fermi surface matching for stable QC compositions.*
  - [327] A. Pochelon et al., “Enhanced confinement in negative-triangularity shaped tokamak plasmas,” *Plasma Physics and Controlled Fusion*, Outstanding Paper Prize, 2026. *Negative-triangularity plasma shaping for improved MHD stability.*
  - [328] Various, “Plasmacoustic Metalayers: Corona Discharge Transducers for Broadband Active Acoustic Metamaterials,” in Proc. EAA/Phononics, 2025. *Non-resonant acoustic metamaterial approach using plasma actuators.*
  - [329] M. Artin, *Algebra*, 2nd ed., Pearson, 2010.
  - [330] D. P. Bartel and J. W. Szostak, “Isolation of new ribozymes from a large pool of random sequences,” *Science*, vol. 261, pp. 1411–1418, 1993.
  - [331] H. Beinert, R. H. Holm, and E. Münck, “Iron-sulfur clusters: Nature’s modular, multipurpose structures,” *Science*, vol. 277, no. 5326, pp. 653–659, 1997.
  - [332] L. Bindi, P. J. Steinhardt, N. Yao, and P. J. Lu, “Natural quasicrystals,” *Science*, vol. 324, no. 5932, pp. 1306–1309, 2009.
  - [333] F. Birch, “Elasticity and constitution of the earth’s interior,” *J. Geophys. Res.*, vol. 57, no. 2, pp. 227–286, 1952.
  - [334] R. Breslow, “On the mechanism of the formose reaction,” *Tetrahedron Letters*, vol. 1, no. 21, pp. 22–26, 1959.
  - [335] G. Brostigen and A. Kjekshus, “Redetermined crystal structure of FeS<sub>2</sub> (pyrite),” *Acta Chem. Scand.*, vol. 23, pp. 2186–2188, 1969.
  - [336] B. K. Burgess and D. J. Lowe, “Mechanism of Molybdenum Nitrogenase,” *Chemical Reviews*, vol. 96, no. 7, pp. 2983–3012, 1996.
  - [337] A. Butlerow, “Bildung einer zuckerartigen Substanz durch Synthese,” *Justus Liebigs Annalen der Chemie*, vol. 120, pp. 295–298, 1861. *Discovery of the formose reaction.*

- [338] R. V. Eck and M. O. Dayhoff, "Evolution of the structure of ferredoxin based on living relics of primitive amino acid sequences," *Science*, vol. 152, no. 3720, pp. 363–366, 1966.
- [339] E. A. Gaucher, S. Govindarajan, and O. K. Ganesh, "Palaeotemperature trend for Precambrian life inferred from resurrected proteins," *Nature*, vol. 451, pp. 704–707, 2008.
- [340] W. Gilbert, "Origin of life: The RNA world," *Nature*, vol. 319, p. 618, 1986.
- [341] G. A. Glatzmaier and P. H. Roberts, "A three-dimensional self-consistent computer simulation of a geomagnetic field reversal," *Nature*, vol. 377, pp. 203–209, 1995.
- [342] D. O. Hall, R. Cammack, and K. K. Rao, "Role for ferredoxins in the origin of life and biological evolution," *Nature*, vol. 233, pp. 136–138, 1971.
- [343] J. E. Hein, E. Tse, and D. G. Blackmond, "A route to enantiopure RNA precursors from nearly racemic starting materials," *Nature Chemistry*, vol. 3, pp. 704–706, 2011. (Key cited as Hein00 in ch.20; year approximate.)
- [344] B. Hille, *Ion Channels of Excitable Membranes*, 3rd ed., Sinauer Associates, 2001.
- [345] H. Hosoya, "Topological Index: A Newly Proposed Quantity Characterizing the Topological Nature of Structural Isomers of Saturated Hydrocarbons," *Bull. Chem. Soc. Japan*, vol. 44, no. 9, pp. 2332–2339, 1971.
- [346] H. Hosoya, "Fibonacci numbers and Fibonacci cubes," *Fibonacci Quarterly*, vol. 14, pp. 173–181, 1976.
- [347] C. Huber and G. Wächtershäuser, "Activated Acetic Acid by Carbon Fixation on (Fe,Ni)S Under Primordial Conditions," *Science*, vol. 276, pp. 245–247, 1997.
- [348] C. Huber and G. Wächtershäuser, "Peptides by activation of amino acids with CO on (Ni,Fe)S surfaces: Implications for the origin of life," *Science*, vol. 281, pp. 670–672, 1998.
- [349] D. T. Johnston, F. Wolfe-Simon, A. Pearson, and A. H. Knoll, "Anoxygenic photosynthesis modulated Proterozoic oxygen and sustained Earth's middle age," *Proc. Natl. Acad. Sci. USA*, vol. 106, pp. 16925–16929, 2009. (Key cited as Johnston01 in ch.20.)
- [350] M. Kohmoto, L. P. Kadanoff, and C. Tang, "Localization problem in one dimension: Mapping and escape," *Phys. Rev. Lett.*, vol. 50, no. 23, pp. 1870–1872, 1983. *Fibonacci chain tight-binding model with Cantor-set energy spectrum.*
- [351] N. Lancaster, "Iron-Sulfur Clusters in Biological Electron Transfer and Sensing," in *Comprehensive Inorganic Chemistry II*, Elsevier, 2013.
- [352] N. Lancaster et al., "X-ray emission spectroscopy evidences a central carbon in the FeMocofactor of nitrogenase," *Science*, vol. 334, pp. 940–943, 2011. (Key cited as Lancaster14 in ch.20.)
- [353] N. Lane, J. F. Allen, and W. Martin, "How did LUCA make a living? Chemiosmosis in the origin of life," *BioEssays*, vol. 32, pp. 271–280, 2010.

- 
- [354] D. La Rue, *Principia Psychedelia*, Spectrality Institute, 2026.
  - [355] W. Martin and M. J. Russell, “On the origins of cells: a hypothesis for the evolutionary transitions from abiotic geochemistry to chemoautotrophic prokaryotes, and from prokaryotes to nucleated cells,” *Philos. Trans. R. Soc. B*, vol. 358, pp. 59–85, 2003.
  - [356] T. M. McCollom and J. S. Seewald, “Carbon isotope composition of organic compounds produced by abiotic synthesis under hydrothermal conditions,” *Earth Planet. Sci. Lett.*, vol. 243, pp. 74–84, 2006. (*Key cited as McCollom99 in ch.20.*)
  - [357] W. F. McDonough and S. -S. Sun, “The composition of the Earth,” *Chem. Geology*, vol. 120, pp. 223–253, 1995.
  - [358] H. Michibata, T. Uyama, and K. Kanamori, “The accumulation and reduction of vanadium by ascidians,” in *Vanadium in Biological Systems*, Springer, pp. 149–169, 2003.
  - [359] H. J. Morowitz, *Beginnings of Cellular Life: Metabolism Recapitulates Biogenesis*, Yale University Press, 1992.
  - [360] J. Oró, “Mechanism of synthesis of adenine from hydrogen cyanide under possible primitive earth conditions,” *Nature*, vol. 191, pp. 1193–1194, 1961.
  - [361] M. A. Pasek and D. S. Lauretta, “Aqueous corrosion of phosphide minerals from iron meteorites: A highly reactive source of prebiotic phosphorus on the surface of the early Earth,” *Astrobiology*, vol. 5, no. 4, pp. 515–535, 2005.
  - [362] M. J. Russell and A. J. Hall, “The emergence of life from iron monosulphide bubbles at a submarine hydrothermal redox and pH front,” *J. Geol. Soc. Lond.*, vol. 154, pp. 377–402, 1997.
  - [363] W. Saenger, *Principles of Nucleic Acid Structure*, Springer-Verlag, 1984.
  - [364] E. Schrödinger, *What is Life? The Physical Aspect of the Living Cell*, Cambridge University Press, 1944. *Alias for schrodinger44.*
  - [365] T. Spatzal, M. Aksoyoglu, L. Zhang, S. L. A. Andrade, E. Schleicher, S. Weber, D. C. Rees, and O. Einsle, “Evidence for interstitial carbon in nitrogenase FeMo cofactor,” *Science*, vol. 334, pp. 940, 2011.
  - [366] J. A. Tarduno et al., “A Hadean to Paleoarchean geodynamo recorded by single zircon crystals,” *Science*, vol. 349, pp. 521–524, 2015.
  - [367] A. P. Tsai and J. Q. Guo, “Stable binary quasicrystals: The Cd-based family,” *Phil. Mag. Lett.*, vol. 81, pp. L123–L128, 2001.
  - [368] F. J. Vine and D. H. Matthews, “Magnetic anomalies over oceanic ridges,” *Nature*, vol. 199, pp. 947–949, 1963.
  - [369] K. L. Von Damm, “Seafloor hydrothermal activity: Black smoker chemistry and chimneys,” *Annual Review of Earth and Planetary Sciences*, vol. 18, pp. 173–204, 1990.

- [370] G. Wächtershäuser, “Before enzymes and templates: Theory of surface metabolism,” *Microbiological Reviews*, vol. 52, pp. 452–484, 1988.
- [371] G. Wächtershäuser, “Groundworks for an evolutionary biochemistry: The iron-sulphur world,” *Prog. Biophys. Mol. Biol.*, vol. 58, pp. 85–201, 1992.
- [372] M. C. Weiss, F. L. Sousa, N. Mrnjavac, S. Neukirchen, M. Roettger, S. Nelson-Sathi, and W. F. Martin, “The physiology and habitat of the last universal common ancestor,” *Nature Microbiology*, vol. 1, art. 16116, 2016.
- [373] P. Dumitrescu, J. Bohnet, J. Gaebler, A. Hankin, D. Hayes, A. Kumar, B. Neyenhuis, R. Vasseur, and A. Potter, “Dynamical topological phase realized in a trapped-ion quantum simulator,” *Nature*, vol. 607, pp. 463–467, 2022. *Experimental observation of a prethermal discrete time crystal on 57 qubits using quasi-periodic (Fibonacci) driving.*
- [374] G. S. Engel, T. R. Calhoun, E. L. Read, T.-K. Ahn, T. Mančal, Y.-C. Cheng, R. E. Blankenship, and G. R. Fleming, “Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems,” *Nature*, vol. 446, pp. 782–786, 2007. *First evidence of quantum coherence in biological energy transfer (FMO complex).*
- [375] S. Hammes-Schiffer and A. A. Stuchebrukhov, “Theory of Coupled Electron and Proton Transfer Reactions,” *Chemical Reviews*, vol. 110, no. 12, pp. 6939–6960, 2010.
- [376] W. Man, M. Megens, P. J. Steinhardt, and P. M. Chaikin, “Experimental measurement of the photonic properties of icosahedral quasicrystals,” *Nature*, vol. 436, pp. 993–996, 2005.
- [377] R. A. Marcus, “On the Theory of Oxidation-Reduction Reactions Involving Electron Transfer. I,” *J. Chem. Phys.*, vol. 24, no. 5, pp. 966–978, 1956. *Nobel Prize in Chemistry 1992.*
- [378] F.-A. Popp, “Coherent photon storage of biological systems”, in *Electromagnetic Bio-Information* (F.-A. Popp, ed.), Urban & Schwarzenberg, München, 1979, pp. 123-149. *Foundational work on ultraweak coherent photon emission from living cells.*
- [379] A. P. Tsai, “Discovery of stable icosahedral quasicrystals: progress in understanding structure and properties,” *Chem. Soc. Rev.*, vol. 42, pp. 5352–5363, 2013. *Review of Tsai-type cluster structures in stable icosahedral QCs.*
- [380] Z. V. Vardeny, A. Nahata, and A. Agrawal, “Optics of photonic quasicrystals,” *Nature Photonics*, vol. 7, pp. 177–187, 2013.
- [381] F. Wilczek, “Quantum Time Crystals,” *Physical Review Letters*, vol. 109, 160401, 2012. *Original proposal for discrete time crystals. Alias for wilczek2012.*
- [382] Frausto da Silva, J. J. R. & Williams, R. J. P. (2001). *The Biological Chemistry of the Elements: The Inorganic Chemistry of Life*. 2nd ed., Oxford University Press. *The definitive reference for biological essentiality of chemical elements.*

- 
- [383] Fukui, M. et al. (2004). “Twenty years of experience in monitoring  $^{41}\text{Ar}$  in a research reactor and decrease of its discharge into the environment.” *Health Phys.* **87**(5), 511–519.
- [384] Ohring, M. (2002). *Materials Science of Thin Films: Deposition and Structure*. 2nd ed., Academic Press. §5.7: Sputtering processes.
- [385] Burton, K. & Krebs, H. A. (1953). “The free-energy changes associated with the individual steps of the tricarboxylic acid cycle.” *Biochem. J.* **54**(1), 94–107.
- [386] The Univalent Foundations Program. *Homotopy Type Theory: Univalent Foundations of Mathematics*. Institute for Advanced Study, 2013.
- [387] Tsai, A.P., Inoue, A. and Masumoto, T. *A Stable Quasicrystal in Al-Cu-Fe System*. Japanese Journal of Applied Physics, 1987.
- [388] Johnson, N.W. *Convex polyhedra with regular faces*. Canadian Journal of Mathematics, 1966.
- Canadian Journal of Mathematics, 1966.
- [389] Russell, W. *The Universal One*. Briege Press, 1926.